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Jing Huang

Wuhan University

Yin Dai

Hefei University

Yulin Jiao

Wuhan University

Lican Kang

Wuhan University

Xiliang Lu

`xllv.math@whu.edu.cn`

Wuhan University

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NONASYMPTOTIC CONVERGENCE ANALYSIS FOR THE TAMED UNADJUSTED STOCHASTIC LANGEVIN ALGORITHM

YIN DAI, JING HUANG, YULING JIAO, LICAN KANG, AND XILIAN LU

ABSTRACT. In this work, we consider sampling from a target distribution π_β characterized by the density function $\pi_\beta(\theta) = e^{-\beta U(\theta)} / \int_{\mathbb{R}^d} e^{-\beta U(\theta)} d\theta$ with $\beta > 0$. It is well-known that the Euler-Maruyama discretization of overdamped Langevin stochastic differential equations (SDEs) exhibits instability when the potentials exhibit superlinear growth. Building upon the approach proposed in [7] for mitigating the impact of superlinear drift coefficients in SDEs, we propose a novel Langevin dynamics-based algorithm, termed the Tamed Unadjusted Stochastic Langevin Algorithm (TUSLA), to address the aforementioned sampling problem and establish rigorous performance guarantees. Specifically, we establish a sharp non-asymptotic convergence guarantee in Kullback–Leibler (KL) divergence with the optimal rate of order one, by combining tools from the logarithmic Sobolev inequality (LSI) and the Fokker–Planck equation. As a direct consequence, we further obtain an $O(\lambda^{1/2})$ convergence rate in both Wasserstein-2 and total variation distances, thereby strengthening and generalizing the best-known results in the current literature. Our theoretical findings are supported by comprehensive high-dimensional experiments.

1. INTRODUCTION

In the past decade, Bayesian inference has gained immense popularity as a fundamental tool across various disciplines such as computational statistics, machine learning and Bayesian inverse problems [3, 10, 23, 28, 40, 46, 49, 54, 59]. Unlike traditional methods that provide point estimates, Bayesian approaches aim to generate samples representing the complete posterior distribution of parameters and potential variables. This sampling-based approach enables the assessment of model uncertainty and mitigates the risk of overfitting [14, 42, 57].

The core of data-driven applications lies in the development of algorithms capable of efficiently sampling from high-dimensional target distributions. In particular, let π_β be a probability distribution on $\mathbb{R}^d$ with a density function (also denoted as π_β) relative to the Lebesgue measure for all $\theta \in \mathbb{R}^d$, defined as follows:

$$\pi_\beta(\theta) := e^{-\beta U(\theta)} / \int_{\mathbb{R}^d} e^{-\beta U(\theta)} d\theta, \quad \text{with} \quad \int_{\mathbb{R}^d} e^{-\beta U(\theta)} d\theta < +\infty,$$

where U is typically a continuously differentiable function. Sampling from such a posterior distribution often involves using methods like the Random Walk Metropolis-Hastings algorithm [6, 1, 12, 19, 34]. However, selecting an appropriate proposal distribution for the Metropolis-Hastings algorithm can be challenging. As a result, discretizing a continuous dynamics has been proposed as an alternative approach, which naturally leads to the target distribution π_β becoming an invariant measure of a stochastic differential equation (SDE). Within this framework, consider the overdamped Langevin SDE associated with π_β , defined as:

$$dZ_t = -h(Z_t) dt + \sqrt{2\beta^{-1}} dB_t, \quad Z_0 = \theta_0, \quad (1)$$

where θ_0 is an $\mathbb{R}^d$ -valued random variable, $h = \nabla U$, $\beta > 0$ is the inverse temperature parameter, and $(B_t)_{t \geq 0}$ is a d -dimensional Brownian motion defined on the filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$, satisfying the usual conditions. It is a well-known result that under mild conditions, the Langevin SDE (1) possesses π_β as its unique invariant measure. The corresponding Euler-Maruyama discretization of the Langevin SDE (1) is given by the following update rule:

$$\theta_{n+1} = \theta_n - \lambda_{n+1} h(\theta_n) + \sqrt{2\lambda_{n+1}\beta^{-1}} \xi_{n+1},$$

where $(\xi_n)_{n \geq 1}$ is an independent and identically distributed (i.i.d.) sequence of d -dimensional standard Gaussian random variables, and $(\lambda_n)_{n \geq 1}$ is a sequence of step sizes that can be kept constant or chosen

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to decrease towards 0. It is important to note that the numerical scheme obtained by using the Euler-Maruyama method yields the unadjusted Langevin algorithm (ULA), also known as Langevin Monte Carlo (LMC), which has been extensively studied in the literature. However, it has been well-established that ULA is unstable when the gradient h of the potential function U exhibits super-linear growth, i.e., $\liminf_{|\theta| \rightarrow \infty} |h(\theta)|/|\theta| = +\infty$ [51, Theorem 3.2], [38, 25]. For example, in [38, Lemma 6.3], it is shown that considering the SDE (1) in one dimension with $U(\theta) = |\theta|^4/4$ along with the associated Euler discretization (1), if $\mathbb{E}[|\theta_0|^2] \geq 2/\lambda$ for any $\lambda > 0$, then $\lim_{n \rightarrow +\infty} \mathbb{E}[|\theta_n|^2] = +\infty$. Moreover, the discrete-time Markov chain $(\theta_n)_{n \geq 0}$ generated by ULA diverges towards infinity in finite time with positive probability.

To address the issue of divergence in the case of super-linearly growing gradients, a technique known as “taming” has been introduced in recent literature [7, 26, 33, 32, 44, 52]. This technique involves dividing the gradient h by an appropriately chosen taming factor in the numerical scheme, aiming to control the super-linear growth of h and improve the stability of the algorithm. Based on this idea, to sample the target distribution π_β , one approach is to consider the tamed unadjusted Langevin algorithm (TULA) introduced in [7], which is given by

$$\theta_0^\lambda = \theta_0, \quad \theta_{n+1}^\lambda = \theta_n^\lambda - \lambda G_\lambda(\theta_n^\lambda) + \sqrt{2\lambda} \xi_{k+1}, \quad n \in \mathbb{N}, \quad (2)$$

where the tamed gradient $G_\lambda(\theta)$ is defined as

$$G_\lambda(\theta) := \frac{\nabla U(\theta)}{1 + \lambda |\nabla U(\theta)|} \quad \text{and} \quad G_{\lambda,c}(\theta) := \left(\frac{\partial_i U(\theta)}{1 + \lambda |\partial_i U(\theta)|} \right)_{i \in \{1, \dots, d\}}$$

for all $\theta \in \mathbb{R}^d$. In particular, regardless of whether the gradient U is bounded or not, the tamed gradient G_λ or $G_{\lambda,c}$ is always bounded and satisfies that G_λ or $G_{\lambda,c}$ converges to $h = \nabla U$ when λ tends to zero. Since the classical Langevin Monte Carlo is guaranteed to converge if the gradient U satisfies the global Lipschitz continuity. Therefore, we conjecture that if the tamed gradient h_λ in (5) satisfies the global Lipschitz or linear growth condition, this algorithm (2) with the modified gradient h_λ in (5) should converge faster than TULA, or at the very least not slower than it. Constructing the modified gradient h_λ to satisfy the linear growth condition has been applied to the study of numerical solutions of stochastic differential equations as well as to solving nonconvex stochastic optimization problems that satisfy super-linearly growing and discontinuous stochastic gradients [33, 32, 52, 53].

1.1. Literature review. The literature on the convergence analysis of the Unadjusted Langevin Algorithm (ULA) and its variants has witnessed significant progress in recent years. The initial theoretical guarantees for ULA convergence were established under the assumptions of strongly convex and gradient Lipschitz potentials. Subsequent research efforts have relaxed these assumptions and explored various scenarios.

For the case of strongly convex and gradient Lipschitz functions U , convergence results in Wasserstein distance and total variation distance have been established [15, 17, 18]. Moreover, for log-concave distributions, nonasymptotic convergence rates of ULA have been thoroughly studied [16, 17, 18]. The conditions of strongly convexity have been relaxed to either convexity at infinity or dissipativity assumptions. Under the condition of convexity at infinity, convergence results in Wasserstein-1 distance have been obtained using the reflection coupling technique [11, 36]. Furthermore, [22] extended the nonasymptotic analysis of overdamped Langevin SDE to a broader class of diffusions. Under the dissipativity assumption, [50] established a nonasymptotic estimate in Wasserstein-2 distance, while [58] enhanced the convergence rate through direct analysis of the ergodicity of the overdamped Langevin Monte Carlo.

In addition to the global Lipschitz continuity of the gradient of the potential function, several studies have extended the analysis to settings without global Lipschitz continuity. Two types of settings have been considered: (1) potential functions with Hölder continuous gradients [8, 13, 21, 45]; and (2) potential functions with locally Lipschitz gradients [7, 31, 33, 32].

For weakly smooth (possibly non-smooth) potential functions with Hölder continuous gradients, [8] introduced the perturbed unadjusted Langevin algorithm (PULA) for non-smooth potentials to approximate the target distribution. The main idea of PULA is to add Gaussian smoothing of potential functions used in [43]; the expectation of non-smooth convex potentials with inputs perturbed by Gaussian random vectors is smoother than the potential themselves. However, their analysis was limited to convex potential functions, while many practical applications involve non-convex settings. [21] proposed a beautiful result using tail growth for weakly smooth and weakly dissipative potentials,

leveraging degenerated convexity and modified logarithmic Sobolev inequalities. Building on this approach, [45] relaxed the conditions by considering mixtures of weakly smooth potentials that satisfy weakly dissipative, achieved convergence guarantees under Poincaré inequality or degenerated convexity outside a ball. In addition, [13] analyzed the convergence with respect to the Rényi divergence under either Latała-Oleszkiewicz inequalities or modified logarithmic Sobolev inequalities.

Recent advances in Langevin-type algorithms have made significant progress in addressing instability arising in non-convex and super-linear settings. For gradients that grow super-linearly, the Tamed Unadjusted Langevin Algorithm (TULA) was proposed in [7], offering non-asymptotic convergence guarantees in both total variation and Wasserstein-2 distances by relaxing the global Lipschitz assumption to a polynomial Lipschitz condition. Despite this improvement, convergence in Wasserstein-2 distance still requires an additional strong convexity assumption on the potential function U . In [4], Wasserstein-2 contraction bounds are established for modified Euler schemes, though stability under high-noise regimes demands large diffusion coefficients, which may limit practical applicability. In contrast, [5] demonstrates geometric ergodicity for SDEs with super-linear drifts, leveraging partial dissipativity and uniform ellipticity to obtain exponential convergence rate in Wasserstein-1 distance. Additionally, projected Langevin methods, such as those in [48], provide total variation convergence guarantees for constrained non-convex sampling problems, highlighting the role of domain constraints in stabilizing the dynamics. Collectively, these works illustrate diverse stabilization strategies—gradient taming, enhanced diffusivity, dissipativity, and domain projection—each with its own strengths and limitations.

To address the instability of the stochastic gradient Langevin dynamics (SGLD) algorithm when applied to objective functions with highly non-linear stochastic gradients, [33] analyzed the nonasymptotic properties of the Tamed Unadjusted Stochastic Langevin Algorithm (TUSLA). They considered locally Lipschitz continuity of the gradient in a non-convex setting and obtained nonasymptotic convergence results in Wasserstein-1 and Wasserstein-2 distances, with convergence rates of $1/2$ and $1/4$, respectively. [31] extended the analysis to the case of discontinuous stochastic gradients with locally Lipschitz in average condition, obtaining the same convergence rates as [33] for Wasserstein-1 and Wasserstein-2 distances. Additionally, [32] proposed the extended Tamed Hybrid ε -Order Polygonal Unadjusted Langevin algorithm (e-TH ε O POULA) for optimization problems with discontinuous stochastic gradients. The e-TH ε O POULA algorithm addresses exploding and vanishing gradient problems through a combination of a component-wise taming function and a component-wise boosting function. Nonasymptotic bounds for e-TH ε O POULA were established in Wasserstein-1 and Wasserstein-2 distances under the conditions of locally Lipschitz in average stochastic gradients and a convexity at infinity condition.

1.2. Our contributions. In this paper, we present a novel variant of the Unadjusted Langevin Algorithm (ULA) called the Tamed Unadjusted Stochastic Langevin Algorithm (TUSLA) in (4). We establish explicit nonasymptotic upper bounds for the convergence of TUSLA to the target distribution π_β , measured in both the total variation norm and the Wasserstein-2 distance, under a general “convex at infinity” condition on the potential function U . This significantly relaxes the global Lipschitz or strong convexity assumptions commonly required in previous works such as [7].

More specifically, our results demonstrate that the discrepancy between the law of TUSLA and the target distribution π_β , measured in Wasserstein-2 and total variation distances, converges at a rate of $O(\lambda^{1/2})$ with respect to the step size λ , improving upon the $O(\lambda^{1/4})$ rates obtained in [31, Corollary 2.9], [60, Corollary 2.5], [50, Proposition 10] and [32, 44]. Importantly, we provide explicit bounds with polynomial dependence on both the step size and the number of iterations, and our analysis applies to gradients with super-linear growth, beyond the standard globally Lipschitz setting. To establish these non-asymptotic convergence guarantees, we leverage KL divergence to quantify differences between probability measures with polynomial tails, combined with functional inequality techniques, thereby generalizing the approach of [48, 54] to settings where ∇U satisfies a polynomial growth condition rather than global Lipschitz continuity. To the best of our knowledge, this is the first work providing explicit KL divergence rates for TUSLA under such general conditions.

Our theoretical findings are further supported by extensive high-dimensional numerical experiments, which demonstrate that TUSLA (4) not only achieves the predicted convergence rates but also outperforms TULA (2) and other related algorithms in practical scenarios, especially when the initial point is far from the target distribution.

In summary, our contributions are as follows:

- We derive explicit non-asymptotic convergence rates of order $O(\lambda^{1/2})$ for TUSLA in both Wasserstein-2 and total variation distances, under “convex at infinity” and polynomial growth conditions on the potential. These results generalize and improve upon existing bounds in [7, 31, 60, 50, 32, 44], which typically achieve only $O(\lambda^{1/4})$ rates or require stronger structural assumptions.
- We extend the functional inequality framework to handle super-linear gradient growth, providing explicit estimates for the constant in the logarithmic Sobolev inequality, which enables precise non-asymptotic analysis of TUSLA.
- Through extensive numerical experiments, we demonstrate that TUSLA is robust and efficient in high-dimensional sampling problems, particularly in scenarios with poor initializations, outperforming TULA in both stability and convergence speed.

1.3. Organization of the paper and some notations. The paper is organized as follows. In Section 2, we present the precise statements of the assumptions and the main results. Section 3 provides an overview of the methodology used to establish our theoretical bounds. Some high-dimensional numerical examples which supported our theoretical bounds are given in Section 4. Finally, the proofs of main results are deferred to Appendix A.1-A.9.

We introduce some notations that will be frequently used in the sequel. Let $(\Omega, \mathcal{F}, \mathbb{P})$ denote a probability space, where Ω is the sample space, $\mathcal{F}$ is the sigma-algebra of events, and $\mathbb{P}$ is the probability measure. The expectation of a random variable Z is denoted by $\mathbb{E}[Z]$. For $1 \leq p < \infty$, the space of p -integrable real-valued random variables is denoted by L^p .

Let d be a fixed integer greater than or equal to 1. For an $\mathbb{R}^d$ -valued random variable Z , its law on the Borel sigma-algebra $\mathcal{B}(\mathbb{R}^d)$ is denoted by $\mathcal{L}(Z)$. The Euclidean scalar product is represented by $\langle \cdot, \cdot \rangle$, and $|\cdot|$ denotes the corresponding norm. Additionally, for $a \in \mathbb{R}_+$, $\lfloor a \rfloor$ and $\lceil a \rceil$ respectively denote the floor and ceiling functions evaluated at a .

Let $k \in \mathbb{N}$, $m_0, m_1 \in \mathbb{N}$, and Ω_0, Ω_1 be two open sets in $\mathbb{R}^{m_0}$ and $\mathbb{R}^{m_1}$, respectively. The notation $C^k(\Omega_0; \Omega_1)$ refers to the set of k -times continuously differentiable functions mapping Ω_0 to Ω_1 . For $f \in C^2(\mathbb{R}^d; \mathbb{R})$, ∇f , $\nabla^2 f$, and Δf denote the gradient, Hessian, and Laplacian of f , respectively. For $k \in \mathbb{N}$ and $f \in C^k(\mathbb{R}^d; \mathbb{R})$, $D^i f$ represents the i -th order derivative of f for $i \in \{0, 1, \dots, k\}$. For $x \in \mathbb{R}^d$ and $i \in \{1, \dots, k\}$, the notation $|D^i f(x)|$ is defined as follows:

$$|D^0 f(x)| = |f(x)|, \quad |D^2 f(x)| = \sup_{u \in \mathbb{S}^{d-1}} |\nabla^2 f(x)u|, \quad |D^i f(x)| = \sup_{u_1, \dots, u_i \in \mathbb{S}^{d-1}} |D^i f(x)[u_1, \dots, u_i]|.$$

For any $m, n \in \mathbb{N}$, define

$$C_{\text{poly}}(\mathbb{R}^m; \mathbb{R}^n) = \{f \in C(\mathbb{R}^m; \mathbb{R}^n) \mid \exists C_q, q \geq 0 \text{ such that } |f(x)| \leq C_q(1 + |x|^q) \text{ for all } x \in \mathbb{R}^m\}.$$

Let $\mathcal{P}(\mathbb{R}^d)$ represent the set of probability measures on $\mathcal{B}(\mathbb{R}^d)$. For $\mu \in \mathcal{P}(\mathbb{R}^d)$ and a non-negative measurable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, we denote $\mu(f) = \int_{\mathbb{R}^d} f(\theta) \mu(d\theta)$. Given a Markov kernel P on $\mathbb{R}^d$ and a function f integrable under $P(x, \cdot)$, we denote $Pf(x) = \int_{\mathbb{R}^d} f(y) P(x, dy)$. The total variation norm between μ and ν is defined as:

$$\|\mu - \nu\|_{\text{TV}} := \sup_{|f| \leq 1} |\mu(f) - \nu(f)|.$$

Let μ and ν be two probability measures on a state space Ω with a given sigma-algebra. If μ is absolutely continuous with respect to ν , we denote the Radon-Nikodym derivative of μ with respect to ν as $\frac{d\mu}{d\nu}$. The Kullback-Leibler divergence of μ with respect to ν is defined as:

$$\text{KL}(\mu \mid \nu) = \int_{\Omega} \left(\frac{d\mu}{d\nu} \right) \log \left(\frac{d\mu}{d\nu} \right) d\nu.$$

For any $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$, $\mathcal{D}(\mu, \nu)$ denotes the set of probability measures ζ on $\mathcal{B}(\mathbb{R}^{2d})$ such that their respective marginals are μ and ν . For two Borel probability measures μ and ν defined on $\mathbb{R}^d$ with finite p -th moments, the Wasserstein distance of order $p \geq 1$ is defined as:

$$W_p(\mu, \nu) := \left(\inf_{\zeta \in \mathcal{D}(\mu, \nu)} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} |\theta - \theta'|^p \zeta(d\theta, d\theta') \right)^{1/p}.$$

2. ASSUMPTIONS AND MAIN RESULTS

Let $U : \mathbb{R}^d \rightarrow \mathbb{R}$ be a twice continuously differentiable function, and denote by $h := \nabla U$ its gradient. Additionally, for any $\beta > 0$, we define the probability measure $\pi_\beta(A)$ for $A \in \mathcal{B}(\mathbb{R}^d)$ as follows:

$$\pi_\beta(A) := \frac{\int_A e^{-\beta U(\theta)} d\theta}{\int_{\mathbb{R}^d} e^{-\beta U(\theta)} d\theta}, \quad (3)$$

where we assume that $\int_{\mathbb{R}^d} e^{-\beta U(\theta)} d\theta < \infty$.

In this paper, we propose the tamed unadjusted stochastic Langevin algorithm (TUSLA), which is given by the following iteration:

$$\theta_0^\lambda := \theta_0, \quad \theta_{n+1}^\lambda = \theta_n^\lambda - \lambda h_\lambda(\theta_n^\lambda) + \sqrt{2\lambda\beta^{-1}}\xi_{n+1}, \quad n \in \mathbb{N}, \quad (4)$$

where $\lambda > 0$ is the step size, and $(\xi_n)_{n \in \mathbb{N}}$ is a sequence of independent standard d -dimensional normal random variables independent of the $\mathbb{R}^d$ -valued random variable θ_0 . Moreover, for any $\theta \in \mathbb{R}^d$, the tamed gradient h_λ is defined as follows:

$$h_\lambda(\theta) := \frac{h(\theta)}{1 + \sqrt{\lambda}|\theta|^r}. \quad (5)$$

TUSLA generates samples from the target distribution π_β by simulating a Markov chain using the update rule in (4). It incorporates a tamed gradient h_λ to control the growth of the drift in overdamped Langevin diffusion (1) and enhance exploration of the target distribution. Specifically, $h_\lambda(\theta)$ in (5) scales the gradient $h(\theta)$ based on the magnitude of θ and the step size λ , with the power r determining its growth behavior. This taming mechanism ensures numerical stability and prevents divergence even for large $|\theta|$. The step size λ sets the size of each update, while the inverse temperature β controls the concentration of the target distribution. At each iteration, the state θ_{n+1}^λ is updated from θ_n^λ with a random perturbation $\sqrt{2\lambda\beta^{-1}}\xi_{n+1}$, where ξ_{n+1} is drawn from a standard normal distribution, enabling effective exploration of the space. Therefore, TUSLA provides a stable and efficient approach for sampling from π_β .

2.1. Definition and functional inequalities. In this subsection, we review the definition of Poincaré inequality, Log-Sobolev inequality, defective Log-Sobolev inequality, Talagrand inequality and the Otto-Villani theorem in [47, Theorem 1].

Definition 2.1 (Poincaré Inequality). *A probability measure $\pi \in \mathcal{P}(\mathbb{R}^d)$ satisfies the Poincaré inequality with a constant $C > 0$, denoted $\text{PI}(C)$, if for all smooth $g : \mathbb{R}^d \rightarrow \mathbb{R}$,*

$$\text{Var}_\pi(g) := \int_{\mathbb{R}^d} \left(g - \int_{\mathbb{R}^d} g d\pi \right)^2 d\pi \leq \frac{1}{C} \int_{\mathbb{R}^d} |\nabla g|^2 d\pi.$$

Given two measures $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$, then the relative Fisher information of μ with respect to ν is defined as

$$I_\nu(\mu) := \int_{\mathbb{R}^d} \left| \nabla \log \frac{\mu(x)}{\nu(x)} \right|^2 \mu(dx).$$

Definition 2.2 (Log-Sobolev Inequality). *A probability measure $\pi \in \mathcal{P}(\mathbb{R}^d)$ satisfies the logarithmic Sobolev inequality with a constant $C > 0$, denoted $\text{LSI}(C)$, if for all smooth $g : \mathbb{R}^d \rightarrow \mathbb{R}$ with $\pi(g^2) < \infty$,*

$$\int_{\mathbb{R}^d} g^2 \log g^2 d\pi - \left(\int_{\mathbb{R}^d} g^2 d\pi \right) \log \left(\int_{\mathbb{R}^d} g^2 d\pi \right) \leq \frac{2}{C} \int_{\mathbb{R}^d} |\nabla g|^2 d\pi.$$

Remark 2.3. LSI is equivalent to the following relation between Kullback-Leibler divergence and relative Fisher information for all $\mu \in \mathcal{P}(\mathbb{R}^d)$:

$$\text{KL}(\mu \mid \pi) \leq \left(\frac{1}{2C} \right) I_\pi(\mu) = \frac{1}{2C} \int_{\mathbb{R}^d} \left| \nabla \log \frac{\mu(x)}{\pi(x)} \right|^2 \mu(dx).$$

Definition 2.4 (Defective Log-Sobolev Inequality). *A probability measure $\pi \in \mathcal{P}(\mathbb{R}^d)$ satisfies the defective Log-Sobolev inequality with constants $C > 0, D \geq 0$, denoted $\text{LSI}(C, D)$, if for all smooth $g : \mathbb{R}^d \rightarrow \mathbb{R}$,*

$$\text{Ent}_\pi(g) := \int_{\mathbb{R}^d} g \log g d\pi - \left(\int_{\mathbb{R}^d} g d\pi \right) \log \left(\int_{\mathbb{R}^d} g d\pi \right) \leq \frac{1}{2C} \int_{\mathbb{R}^d} \frac{|\nabla g|^2}{g} d\pi + D \int_{\mathbb{R}^d} g d\pi.$$

Definition 2.5 (Talagrand Inequality). *A probability measure $\pi \in \mathcal{P}(\mathbb{R}^d)$ satisfies the Transportation-entropy (or Talagrand) inequality with a constant $C > 0$, if for all probability measure μ ,*

$$\frac{C}{2} \cdot W_2(\mu, \pi)^2 \leq \text{KL}(\mu \mid \pi).$$

Theorem 2.6 (Otto-Villani). *Assume that a probability measure π satisfies $\text{LSI}(C^\dagger)$. Then it also satisfies the Talagrand inequality with the same constant $C^\dagger > 0$.*

2.2. Assumptions. In this subsection, we present the conditions required to establish the main results. Let's start by introducing the following assumptions.

Assumption 1. *The initial random variable θ_0 has $4(1+r)$ -order moment, i.e., $\mathbb{E}[|\theta_0|^{4(1+r)}] < \infty$.*

Now, let's introduce another assumption that provides conditions on the local Lipschitz continuity and polynomial growth of $h(\cdot)$.

Assumption 2. *There exist constants $L > 0, r \in \mathbb{N}$ such that, for all $\theta, \theta' \in \mathbb{R}^d$,*

$$|h(\theta) - h(\theta')| \leq L(1 + |\theta| + |\theta'|)^r |\theta - \theta'|.$$

Furthermore, there exists a constant $K > 0$ such that, for all $\theta \in \mathbb{R}^d$,

$$|h(\theta)| \leq K(1 + |\theta|^{r+1}).$$

Remark 2.7. *In general, we assume that r is a positive integer. However, Assumption 2 can also cover the case when $r = 0$, in which the gradient function $h(\cdot)$ is globally Lipschitz continuous.*

Remark 2.8. *From the locally Lipschitz condition of h in Assumption 2, we can deduce a polynomial growth condition for h . Specifically, let $K = \max\{L2^{r-1} + |h(0)|, L2^r\}$. The choice of K in Assumption 2 is motivated by the aim to optimize the recursion on the step-size, which is proportional to the reciprocal of K . For example, consider the case of $p \in [1, +\infty) \cap \mathbb{N}$ and $h(\theta) = \theta(|\theta|^{2p} - 1)$ for all $\theta \in \mathbb{R}^d$. Using the inequality*

$$||\theta|^{2p} - |\theta'|^{2p}| \leq 2p(|\theta| + |\theta'|)^{2p-1} |\theta - \theta'|,$$

we obtain the inequality

$$|h(\theta) - h(\theta')| = |\theta(|\theta|^{2p} - 1) - \theta'(|\theta'|^{2p} - 1)| \leq 2p(1 + |\theta| + |\theta'|)^{2p} |\theta - \theta'|.$$

Hence, Assumption 2 is satisfied with $L = 2p$ and $r = 2p$, which further implies

$$|h(\theta)| = |\theta(|\theta|^{2p} - 1)| \leq K(1 + |\theta|^{2p+1}),$$

where $K := (2p)4^p$. However, if we directly use the expression of h , we obtain $|h(\theta)| = |\theta(|\theta|^{2p} - 1)| \leq K(1 + |\theta|^{2p+1})$ with $K = 2$. This is the reason why we choose to impose a separate polynomial growth condition on h .

In Assumption 3, we impose a local convexity at infinity condition on the gradient h . Let's further analyze and discuss this assumption.

Assumption 3. *The gradient h satisfies the following assumptions:*

(i) *If $r > 0$, then there exist constants $a, b > 0$, and $0 \leq \bar{r} < r$ such that, for all $\theta, \theta' \in \mathbb{R}^d$,*

$$\langle \theta - \theta', h(\theta) - h(\theta') \rangle \geq a|\theta - \theta'|^2(|\theta|^r + |\theta'|^r) - b|\theta - \theta'|^2(|\theta|^{\bar{r}} + |\theta'|^{\bar{r}}). \quad (6)$$

(ii) *If $r = 0$, then there exist constants $\bar{a}, \bar{b}$ such that, for any $\theta \in \mathbb{R}^d$,*

$$\langle \theta, h(\theta) \rangle \geq \bar{a}|\theta|^2 - \bar{b}. \quad (7)$$

To better understand Assumption 3, let's consider the condition:

$$\langle \theta - \theta', h(\theta) - h(\theta') \rangle \geq a|\theta - \theta'|^2(|\theta|^r + |\theta'|^r). \quad (8)$$

If $r > 0$, Assumption 3 can be seen as a ‘‘convexity at infinity’’ condition. In particular, when $|\theta|$ and $|\theta'|$ are large enough, the first term on the right-hand side of condition (6) dominates, and the condition essentially behaves like condition (8). It is important to note that condition (8) is stronger than the usual strong convexity condition (H3) in [7], as the strongly convex constant in condition (8) depends on θ and θ' .

However, when $|\theta|$ and $|\theta'|$ are sufficiently small, the second term on the right-hand side of condition (6) dominates, and $\langle \theta - \theta', h(\theta) - h(\theta') \rangle$ may be strictly less than 0. This means that U may not satisfy convexity on domains where θ and θ' are small enough. In other words, the convexity assumption at infinity may lead to the non-convexity of the potential function U . However, this trade-off is necessary to impose stronger assumptions on U than strong convexity when $|\theta|$ and $|\theta'|$ are large enough.

Example 2.9. Consider the gradient of a double-well potential $h(\theta) = (|\theta|^2 - 1)\theta$. On one hand, we can derive:

$$\langle \theta - \theta', h(\theta) - h(\theta') \rangle \geq \frac{1}{2} (|\theta|^2 + |\theta'|^2 - 2) |\theta - \theta'|^2$$

for any $\theta, \theta' \in \mathbb{R}^d$. Thus, Assumption 3 holds with $a = \frac{1}{2}$, $b = 1$, $r = 2$ and $\bar{r} = 0$. On the other hand, for all $|\theta|, |\theta'| \geq \sqrt{2}$, we also have:

$$\langle \theta - \theta', h(\theta) - h(\theta') \rangle \geq |\theta - \theta'|^2.$$

Therefore, the function h is not convex on $\mathbb{R}^d$ due to its behavior when θ and θ' are near 0. However, h is a strongly convex function on subdomains where $|\theta|$ and $|\theta'|$ are both greater than or equal to $\sqrt{2}$.

Under Assumptions 2-3, we can establish a dissipativity condition for the drift function h . This result is presented in Remark 2.10, along with an explicit expression for the constants involved.

Remark 2.10. Under Assumptions 2-3, there exist positive constants a^* and b^* such that, for all $\theta \in \mathbb{R}^d$,

$$\langle \theta, h(\theta) \rangle \geq a^* |\theta|^{r+2} - b^*. \quad (9)$$

The constants a^* and b^* are explicitly given by

$$a^* = \frac{a}{2} \mathbb{1}_{r>0} + \bar{a} \mathbb{1}_{r=0},$$

$$b^* = \left\{ \left(\frac{a}{2} + b \right) D^{2+\bar{r}} + \frac{K^2}{2a} \right\} \mathbb{1}_{r>0} + \bar{b} \mathbb{1}_{r=0},$$

where $D = \max \left\{ \left(\frac{4b}{a} \right)^{1/(r-\bar{r})}, 2^{1/r} \right\}$ and $\mathbb{1}$ denotes the indicator function. Additionally, we have the following inequality for all $\theta \in \mathbb{R}^d$:

$$\langle \theta, h(\theta) \rangle \geq a^* |\theta|^2 - b^\sharp, \quad (10)$$

where

$$b^\sharp = (b^* + a^*) \mathbb{1}_{r>0} + \bar{b} \mathbb{1}_{r=0}.$$

The proof of this result can be found in the Appendix A.2.

2.3. Main results. Let $(P_t)_{t \geq 0}$ be the semigroup associated with the Langevin SDE (1). For any $x \in \mathbb{R}^d$ and $A \in \mathcal{B}(\mathbb{R}^d)$, we have $P_t(x, A) = \mathbb{E} [\mathbb{1}_A(Z_t) | Z_0 = x]$. The infinitesimal generator $\mathcal{A}$ associated with SDE (1) is defined as

$$\mathcal{A}g(x) = -\langle h(x), \nabla g(x) \rangle + \Delta g(x) = -\langle \nabla U(x), \nabla g(x) \rangle + \Delta g(x), \quad (11)$$

for any $x \in \mathbb{R}^d$, where $\Delta g(x)$ represents the Laplacian of g at x . Define the maximum step-size restriction $\lambda_{\max}$ of TUSLA (4) as

$$\lambda_{\max} := \min \left\{ 1, \frac{(a^*)^2}{4K^4}, \frac{1}{(a^*)^2} \right\}, \quad (12)$$

where K, a^* are constants explicitly given in Assumption 2 and (9) in Remark 2.10. And let π_n be the law of n -th iterate of TUSLA (4). Based on these definitions, we can state the following theorem.

Theorem 2.11. Assume that Assumptions 1, 2 and 3 are satisfied. Let $\pi_0 := \mathcal{L}(\theta_0)$ and $\text{KL}(\pi_0 | \pi_\beta) < \infty$. Then, for any $n \in \mathbb{N}$, $0 < \lambda \leq \lambda_{\max}$ with $\lambda_{\max}$ given in (12),

$$\text{KL}(\pi_n | \pi_\beta) \leq \exp \left\{ -n\lambda \left(\frac{3C_{\text{LSI}}}{2\beta} \right) \right\} \text{KL}(\pi_0 | \pi_\beta) + \left\{ \frac{4\beta^2(C_1^\dagger + C_2^\dagger)}{3C_{\text{LSI}}} \right\} \lambda,$$

where constants $C_1^\dagger, C_2^\dagger, C_{\text{LSI}}$ are given in Lemma 3.11, Proposition 3.2 and Corollary 3.8. In addition, for any $\varepsilon \in (0, 1)$ and $\lambda \in (0, \lambda_{\max})$, one also obtains

$$\text{KL}(\pi_n | \pi_\beta) \leq \varepsilon$$

provided

$$n \geq \left\{ \frac{16\beta^3(C_1^\dagger + C_2^\dagger)}{(3C_{\text{LSI}})^2\varepsilon} \right\} \log \left(\frac{2\text{KL}(\pi_0 \mid \pi_\beta)}{\varepsilon} \right) \quad \text{and} \quad \lambda \leq \left\{ \frac{3C_{\text{LSI}}}{8(C_1^\dagger + C_2^\dagger)} \right\} \beta^{-2}\varepsilon.$$

Remark 2.12. The Kullback-Leibler divergence of the initial condition θ_0 relative to the target distribution π_β is finite can be satisfied, for example, by taking the Gaussian random variable $\theta_0 \sim N(0, \mathbf{I}_d)$, then one obtains

$$\text{KL}(\pi_0 \mid \pi_\beta) \leq \frac{d}{2} \log \left(\frac{1}{2\pi e} \right) + \beta \left\{ U(0) + \frac{K(2d)^{\frac{2+r}{2}} \Gamma(\frac{3+r}{2})}{\sqrt{\pi}(2+r)} + Kd\sqrt{\frac{2}{\pi}} \right\} < \infty,$$

where $\Gamma(\cdot)$ denotes the Gamma function.

Proof. See Appendix A.2. □

Corollary 2.13. Under the same assumptions as Theorem 2.11, then for any $n \in \mathbb{N}$, $0 < \lambda \leq \lambda_{\max}$ with $\lambda_{\max}$ given in (12),

$$W_2(\mathcal{L}(\theta_n^\lambda), \pi_\beta) \leq \sqrt{\frac{2}{C_{\text{LSI}}}} \exp \left\{ -n\lambda \left(\frac{3C_{\text{LSI}}}{4\beta} \right) \right\} \sqrt{\text{KL}(\pi_0 \mid \pi_\beta)} + \left\{ \frac{2\beta\sqrt{2(C_1^\dagger + C_2^\dagger)}}{\sqrt{3}C_{\text{LSI}}} \right\} \lambda^{\frac{1}{2}},$$

where constants $C_1^\dagger, C_2^\dagger, C_{\text{LSI}}$ are given in Lemma 3.11, Proposition 3.2 and Corollary 3.8. In addition, for any $\varepsilon \in (0, 1)$ and $\lambda \in (0, \lambda_{\max})$, one also obtains

$$W_2(\mathcal{L}(\theta_n^\lambda), \pi_\beta) \leq \varepsilon$$

provided

$$n \geq \left(\frac{32\beta^2}{\varepsilon C_{\text{LSI}}^3} \right) (C_1^\dagger + C_2^\dagger) \log \left\{ \frac{2\sqrt{2\text{KL}(\pi_0 \mid \pi_\beta)}}{\varepsilon\sqrt{C_{\text{LSI}}}} \right\} \quad \text{and} \quad \lambda \leq \left(\frac{\varepsilon}{8\beta} \right) C_{\text{LSI}}^2 (C_1^\dagger + C_2^\dagger)^{-1}.$$

Proof. The proof follows directly by combining two key ingredients: First, since the target distribution π_β satisfies a logarithmic Sobolev inequality with constant $C_{\text{LSI}} > 0$, the Otto-Villani theorem establishes the fundamental relationship between the Wasserstein-2 distance and the Kullback-Leibler divergence:

$$W_2(\mathcal{L}(\theta_n^\lambda), \pi_\beta) \leq \sqrt{(2/C_{\text{LSI}}) \text{KL}(\mathcal{L}(\theta_n^\lambda) \mid \pi_\beta)}.$$

Second, Theorem 2.11 provides explicit control on $\text{KL}(\mathcal{L}(\theta_n^\lambda) \mid \pi_\beta)$. Combining these two results yields the desired convergence in Wasserstein-2 distance, completing the proof. □

Corollary 2.14. Under the same assumptions as Theorem 2.11, then for any $n \in \mathbb{N}$, $0 < \lambda \leq \lambda_{\max}$ with $\lambda_{\max}$ given in (12),

$$\|\mathcal{L}(\theta_n^\lambda) - \pi_\beta\|_{\text{TV}} \leq \sqrt{2} \exp \left\{ -n\lambda \left(\frac{3C_{\text{LSI}}}{4\beta} \right) \right\} \sqrt{\text{KL}(\pi_0 \mid \pi_\beta)} + \left\{ \frac{2\beta\sqrt{2(C_1^\dagger + C_2^\dagger)}}{\sqrt{3}C_{\text{LSI}}} \right\} \lambda^{\frac{1}{2}},$$

where constants $C_1^\dagger, C_2^\dagger, C_{\text{LSI}}$ are given in Lemma 3.11, Proposition 3.2 and Corollary 3.8. In addition, for any $\varepsilon \in (0, 1)$ and $\lambda \in (0, \lambda_{\max})$, one also obtains

$$\|\mathcal{L}(\theta_n^\lambda) - \pi_\beta\|_{\text{TV}} \leq \varepsilon$$

provided

$$n \geq \left(\frac{128\beta^3}{9\varepsilon^2 C_{\text{LSI}}^3} \right) (C_1^\dagger + C_2^\dagger) \log \left(\frac{2\sqrt{2\text{KL}(\pi_0 \mid \pi_\beta)}}{\varepsilon} \right) \quad \text{and} \quad \lambda \leq \left\{ \frac{3C_{\text{LSI}}^2}{32(C_1^\dagger + C_2^\dagger)} \right\} \beta^{-2}\varepsilon^2.$$

Proof. The desired convergence in total variation norm follows directly from an application of Pinsker's inequality, which provides the quantitative bound

$$\|\mathcal{L}(\theta_n^\lambda) - \pi_\beta\|_{\text{TV}} \leq \sqrt{2 \text{KL}(\mathcal{L}(\theta_n^\lambda) \mid \pi_\beta)}$$

coupled with the exponential decay of $\text{KL}(\mathcal{L}(\theta_n^\lambda), \pi_\beta)$ established in Theorem 2.11. □

To study the empirical average $n^{-1} \sum_{k=0}^{n-1} f(\theta_k^\lambda)$ of a Markov chain $(\theta_k^\lambda)_{k \in \mathbb{N}}$, we employ a method used in [7, Theorem 7] and [39, Theorem 5.1], which relies on the Poisson equation. The Poisson equation associated with the infinitesimal generator $\mathcal{A}$ defined in (11) is given by

$$\mathcal{A}f(x) = -(f(x) - \pi_\beta(f)),$$

for all $x \in \mathbb{R}^d$, where $\pi_\beta(f)$ denotes the integral of f with respect to the stationary distribution π_β . To estimate the variance of the empirical mean $n^{-1} \sum_{k=0}^{n-1} f(\theta_k^\lambda)$, we require the following two additional assumptions about the potential function U .

Assumption 4. *The gradient $h(\cdot)$ satisfies the following:*

$$\liminf_{|\theta| \rightarrow +\infty} |h(\theta)| = +\infty, \quad \liminf_{|\theta| \rightarrow +\infty} \frac{\langle \theta, h(\theta) \rangle}{|\theta| \cdot |h(\theta)|} > 0.$$

Assumption 5. *Assume that $U \in C^4(\mathbb{R}^d; \mathbb{R})$, and $|D^i U| \in C_{\text{poly}}(\mathbb{R}^d; \mathbb{R}_+)$ for all $i \in \{1, 2, 3, 4\}$.*

Proposition 2.15. *Assume that Assumptions 2, 3, 4 and 5 hold. Let $f \in C^3(\mathbb{R}^d; \mathbb{R})$ satisfy $|D^i f| \in C_{\text{poly}}(\mathbb{R}^d; \mathbb{R}_+)$ for all $i \in \{0, 1, 2, 3\}$. Let $\lambda_0 > 0$, and $(\theta_k^\lambda)_{k \in \mathbb{N}}$ be the Markov chain defined by (4) with an initial state $\theta_0^\lambda = \theta_0$. There exists a constant $C^* > 0$ such that for all $\lambda \in (0, \lambda_0]$ and $n \in \mathbb{N}$, the following bounds hold:*

$$\left| \mathbb{E} \left[\frac{1}{n} \sum_{k=0}^{n-1} f(\theta_k^\lambda) - \pi_\beta(f) \right] \right| \leq C^* \left(\lambda + \frac{1}{n\lambda} \right)$$

and

$$\mathbb{E} \left\{ \left(\frac{1}{n} \sum_{k=0}^{n-1} f(\theta_k^\lambda) - \pi_\beta(f) \right)^2 \right\} \leq C^* \left(\lambda^2 + \frac{1}{n\lambda} \right).$$

Proof. Assumptions 2, 4 and 5 and Lemma A.1 in Appendix A.3 satisfy all the assumptions in [7, Theorem 7]. Therefore, the desired result is obtained. $\square$

Remark 2.16. *It is worth noting that the convergence rates obtained in [7, Theorem 7] and [39, Theorems 5.1 and 5.2] can be recovered using the current framework.*

3. PROOF OF THE MAIN RESULTS

In the context of studying the Langevin SDE (1), it is well-known that it has a unique strong solution under Assumptions 2 and 3. This result can be found in [37, Theorem 3.6] or see Appendix A.1. We introduce several auxiliary processes which are key for the analysis of the convergence results. Then, the explicit moment bounds are obtained for the auxiliary processes. Finally, we provide detailed explanations of the proofs of the main results.

3.1. Explicit characterization of the Log-Sobolev constant. This subsection establishes explicit bounds for the Log-Sobolev constant of the Gibbs measure $\pi_\beta(d\theta) \propto \exp(-\beta U(\theta)) d\theta$, analyzing its dependence on both the inverse temperature β and dimension d for two cases: under Assumption 2 with $r = 0$ and under Assumption 2 with $r > 0$.

For the first case, we only need consider dissipativity and global Lipschitz continuity, we can obtain the Log-Sobolev constant whose inverse exponential depends on the inverse temperature β . This employs the following results from [50, Proposition 9].

Proposition 3.1 ([50]). *Let Assumption 2 and 3 hold with $r = 0$. Then for all $\beta \geq \frac{2}{\bar{a}}$, probability measure π_β satisfies Log-Sobolev inequality with constant C_{LSI} such that*

$$\frac{1}{C_{\text{LSI}}} \leq \frac{2(\bar{a})^2 + 2L^2}{\beta L(\bar{a})^2} + \frac{1}{\lambda_*} \left\{ \frac{6L(d + \beta)}{\bar{a}} + 2 \right\},$$

where the spectral gap λ_* satisfies

$$\frac{1}{\lambda_*} \leq \frac{1}{\bar{a}(d + \bar{b}\beta)\beta} + \frac{2C_*(d + \bar{b}\beta)}{\beta \bar{a}} \exp \left\{ \frac{2}{\bar{a}}(L + B_*)(\beta \bar{b} + d) + \beta(A_* + B_*) \right\}.$$

Here, $A_* := |U(0)|$, $B_* := |h(0)|$ and C_* is a universal constant independent of gradient h .

This leads to our refined characterization:

Proposition 3.2. *Under the same assumptions as Proposition 3.1, the Log-Sobolev constant C_{LSI} of π_β can be written with constants $C_1^\sharp, C_2^\sharp > 0$ independent of β as follows:*

$$C_{\text{LSI}} = \beta C_1^\sharp \exp \left\{ -\beta C_2^\sharp \right\},$$

where

$$C_1^\sharp := \left\{ \frac{2(\bar{a})^2 + 2L^2}{L(\bar{a})^2} + \left(2 + \frac{6Ld}{\bar{a}} \right) \left(\frac{1}{\bar{a}d} + \frac{2C_\star d}{\bar{a}} e^{\frac{2d}{\bar{a}}(L+B_\star)} \right) \right\}^{-1} = e^{-\mathcal{O}(d)}$$

and

$$C_2^\sharp := \frac{2\bar{b}}{\bar{a}}(L + B_\star) + (A_\star + B_\star + \bar{b} + 1).$$

Proof. See Appendix A.5. □

For the second case, the optimal constant for the Log-Sobolev inequality of probability measure π_β remains an open problem that has not been fully resolved. Addressing this theoretical gap is particularly significant as it enables more precise analysis of convergence rates in sampling algorithms. Based on the theoretical framework of [2, Proposition 5.1.3], we establish a quantitative relationship between the Poincaré inequality and the Defective Log-Sobolev inequality, thereby obtaining for the first time a precise upper bound of the Log-Sobolev constant for π_β .

Specifically, our technical approach consists of two key components: First, building upon the reflection coupling technique developed in [20], we establish a quantitative relationship between the exponential convergence rate of stochastic process $(Z_t)_{t \geq 0}$ defined in (1) under Wasserstein-1 distance and the Poincaré inequality for the invariant measure π_β . This relationship enables us to precisely determine the optimal Poincaré constant for π_β . Second, by employing the Harnack inequality technique proposed in [56], we derive sufficient conditions for π_β to satisfy the Defective Log-Sobolev inequality and provide explicit expressions for the corresponding constant upper bounds. These results serve as a crucial foundation for establishing the Log-Sobolev inequality.

The following proposition establishes that the conjunction of a defective logarithmic Sobolev inequality and a Poincaré inequality guarantees a sharpened logarithmic Sobolev inequality, as shown in [2, Proposition 5.1.3].

Proposition 3.3 ([2]). *Assume the defective logarithmic Sobolev inequality $\text{LSI}(C, D)$ and Poincaré inequality $\text{PI}(H)$ are hold for $\pi \in \mathcal{P}(\mathbb{R}^d)$, then μ satisfies a tight logarithmic Sobolev inequality with constant $H \left(\left(\frac{D}{2} + 1 \right) + \frac{H}{C} \right)^{-1}$.*

Proposition 3.4. *Let Assumptions 1, 2 and 3 with $r > 0$ hold. Let $(Z_t)_{t \geq 0}, (Z'_t)_{t \geq 0}$ be solutions of the Langevin SDE (1) with initial conditions $Z_0 := \theta_0, Z'_0 = \theta'_0$. Then there exist constants $\hat{c}, \dot{c} > 0$ such that for all $t \geq 0$,*

$$W_1(\mathcal{L}(Z_t), \mathcal{L}(Z'_t)) \leq \hat{c} e^{-\dot{c}t} W_1(\mathcal{L}(\theta_0), \mathcal{L}(\theta'_0)).$$

The explicit from the constants $\hat{c}$ and $\dot{c}$ are given by

$$\hat{c} := 2 \exp \left(-\frac{\bar{L} R_\star^2}{8} \right), \quad \dot{c} := 2\beta^{-1} c_\star^{-1}$$

and

$$c_\star := \begin{cases} \frac{2}{3e} \min \left\{ \frac{1}{R_\star^2}, \frac{m}{8} \right\}, & \text{if } L R_\star^2 \leq 8, \\ \left\{ \frac{8\sqrt{2\pi}}{R_\star \sqrt{L}} \left(\frac{1}{m} + \frac{1}{L} \right) \exp \left(\frac{\bar{L} R_\star^2}{8} \right) + \frac{32}{m^2 R_\star^2} \right\}^{-1}, & \text{if } L R_\star^2 \geq 8, \end{cases}$$

where the constants $\bar{L}, m, R_\star$ are defined in Lemma A.2.

Proof. It follows directly by invoking Theorem 1, Corollary 2 and Lemma 1 in [20] and Lemma A.2. □

Proposition 3.5 (Poincaré Inequality). *Suppose that Assumptions 1, 2 and 3 with $r > 0$ hold. Then Gibbs measure π_β satisfies the Poincaré inequality with constant*

$$C_{\text{PI}} := \left(\frac{\hat{c}}{\sqrt{\dot{c}\beta}} \right)^{-2} = \left(\frac{\sqrt{2/c_\star}}{\hat{c}} \right)^2,$$

where constants $\hat{c}$ and $c_\star$ are given by in Proposition 3.4.

Proof. See Appendix A.5. $\square$

Next, we review the Harnack inequality of Langevin SDE (1) in [56], and refer to [41, proposition 10] for specific proof.

Proposition 3.6 (Harnack Inequality). *Let $(P_t)_{t \geq 0}$ be the semigroup associated with the Langevin SDE (1). Suppose there exists a constant $K > 0$ such that for any $x, y \in \mathbb{R}^d$,*

$$\langle x - y, h(x) - h(y) \rangle \geq -K|x - y|.$$

Then, for any $\alpha > 1, t \geq 1$ and bounded measurable function $f \geq 0$,

$$(P_t f(y))^\alpha \leq (P_t f^\alpha)(x) \exp \left\{ \frac{\alpha}{4\beta^{-1}(\alpha-1)} \left(K^2 t + \frac{|x-y|^2}{t} \right) \right\}.$$

In the following, for any $\alpha, \gamma \geq 1$ we write

$$\|f\|_\alpha := \left(\int_{\mathbb{R}^d} |f|^\alpha(x) \pi_\beta(dx) \right)^{1/\alpha}, \quad \|P_t\|_{\alpha \rightarrow \gamma} := \sup\{\|P_t f\|_\gamma : f \in L^\alpha(\pi_\beta), \|f\|_\alpha = 1\},$$

where π_β is the invariant measure of the semi-group $(P_t)_{t \geq 0}$ considered. The semi-group is said to be hypercontractive if there exist $t_0 > 0, 1 < \alpha < \gamma$ such that $\|P_{t_0}\|_{\alpha \rightarrow \gamma} < \infty$.

Proposition 3.7. *Let $\gamma > \alpha > 1$. Under Assumptions 1, 2 and 3 with $r > 0$, define the time threshold*

$$t_0 := \frac{2\gamma}{m(\alpha-1)/\beta}.$$

Then for all $t > t_0$, one obtains

$$\|P_t\|_{\alpha \rightarrow \gamma} \leq (1 + 4d + (2\bar{L} + m)R_\star^2) \exp \left(\frac{\gamma(\bar{L}R_\star)^2\beta}{4(\alpha-1)} t + \frac{1}{8} \max \left\{ \frac{1+4d}{t/t_0-1}, mR_\star^2 \right\} \right).$$

Proof. See Appendix A.6. $\square$

Corollary 3.8 (Log-Sobolev Inequality). *Under the assumption that Proposition 3.7 holds, then Gibbs measure π_β satisfies defective logarithmic Sobolev inequality $\text{LSI}(C, D)$ with constants*

$$C := \frac{m}{72\beta}, \quad D := 6 \log(1 + 4d + (2\bar{L} + m)R_\star^2) + \left(\frac{54}{m} \right) (\bar{L}R_\star\beta)^2 + \frac{3}{4} \max \{1 + 4d, mR_\star^2\}.$$

Moreover, π_β also satisfies logarithmic Sobolev inequality with constant

$$C_{\text{LSI}} := C_{\text{PI}} \left\{ \left(\frac{D}{2} + 1 \right) + \frac{C_{\text{PI}}}{C} \right\}^{-1} = \mathcal{O}(d^{-1} + \beta^{-2}).$$

Proof. For simplicity, setting $\alpha = 2, \gamma = 3, t = 2t_0$ in Proposition 3.7, one obtains

$$\|P_{2t_0}\|_{2 \rightarrow 3} \leq (1 + 4d + (2\bar{L} + m)R_\star^2) \exp \left(\frac{9(\beta\bar{L}R_\star)^2}{m} + \max \left\{ \frac{1+4d}{8}, \frac{mR_\star^2}{8} \right\} \right), \quad (13)$$

where $t_0 := 3\beta/m$. Applying the hypercontractivity bound in (13) to [2, Theorem 5.2.5], we conclude that probability measure π_β satisfies defective logarithmic Sobolev inequality with explicitly constants

$$C := (12t_0)^{-1} = \frac{m}{72\beta}$$

and

$$D := 6 \log(1 + 4d + (2\bar{L} + m)R_\star^2) + \left(\frac{54}{m} \right) (\bar{L}R_\star\beta)^2 + \frac{3}{4} \max \{1 + 4d, mR_\star^2\}.$$

The conclusion of the second part is obtained by combining the defective logarithmic Sobolev inequality $\text{LSI}(C, D)$ with Properties 3.3 and 3.5. This completes the proof. $\square$

3.2. Auxiliary processes. Recall that the overdamped Langevin SDE $(Z_t)_{t \geq 0}$ given by

$$dZ_t = -h(Z_t) dt + \sqrt{2\beta^{-1}} dB_t,$$

with $Z_0 := \theta_0$, where $(B_t)_{t \geq 0}$ is a standard d -dimensional Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$. Denote by $(\mathcal{F}_t)_{t \geq 0}$ the natural filtration of $(B_t)_{t \geq 0}$, which is assumed to be independent of θ_0 . For each $\lambda > 0$, denote by $Z_t^\lambda := Z_{\lambda t}$, $t \in \mathbb{R}_+$, the time-changed Langevin SDE given by

$$dZ_t^\lambda = -\lambda h(Z_t^\lambda) dt + \sqrt{2\lambda\beta^{-1}} dB_t^\lambda, \quad (14)$$

with the initial condition $Z_0^\lambda := \theta_0$, where $B_t^\lambda = B_{\lambda t}/\sqrt{\lambda}$, $t \geq 0$. Note that $(B_t^\lambda)_{t \geq 0}$ is a d -dimensional standard Brownian motion. For each $\lambda > 0$, denote by $(\mathcal{F}_t^\lambda)_{t \geq 0}$ the natural filtration of $(B_t^\lambda)_{t \geq 0}$ with $\mathcal{F}_t^\lambda := \mathcal{F}_{\lambda t}$ for each $t \geq 0$, which is also independent of θ_0 .

Then, define the continuous-time interpolation of TUSLA (4), denoted by $(\bar{\theta}_t^\lambda)_{t \geq 0}$, as

$$d\bar{\theta}_t^\lambda = -\lambda h_\lambda(\bar{\theta}_{[t]}^\lambda) dt + \sqrt{2\lambda\beta^{-1}} dB_t^\lambda \quad (15)$$

with initial condition $\bar{\theta}_0^\lambda := \theta_0$. By construction, one obtains $\mathcal{L}(\bar{\theta}_n^\lambda) = \mathcal{L}(\theta_n^\lambda)$ for all $n \in \mathbb{N}$. That is, the law of the interpolated process coincides with that of TUSLA (4) at all discrete time points. Let π_t denote the marginal distribution of $\bar{\theta}_t^\lambda$. Notably, given $\bar{\theta}_n^\lambda$, the conditional process $\bar{\theta}_t^\lambda$ for $t \in [n, n+1)$ follows a Gaussian distribution with mean $\bar{\theta}_n^\lambda - \lambda(t-n)h_\lambda(\bar{\theta}_n^\lambda)$ and covariance $2\lambda(t-n)\beta^{-1}\mathbf{I}_d$.

This conditional structure implies that the transition density $\pi_{\bar{\theta}_t^\lambda | \bar{\theta}_n^\lambda}(x | y)$ corresponds to the solution of SDE (15) with constant drift $-\lambda h_\lambda(y)$ and initial condition $\bar{\theta}_n^\lambda = y \in \mathbb{R}^d$. Consequently, it satisfies the Fokker-Planck equation:

$$\partial_t \pi_{\bar{\theta}_t^\lambda | \bar{\theta}_n^\lambda}(x | y) = \lambda \left\{ \nabla \cdot \left(\pi_{\bar{\theta}_t^\lambda | \bar{\theta}_n^\lambda}(x | y) h_\lambda(y) \right) + \beta^{-1} \Delta_x \pi_{\bar{\theta}_t^\lambda | \bar{\theta}_n^\lambda}(x | y) \right\}. \quad (16)$$

3.3. Moment bounds. The following lemmas provide moment estimates for the sequence $(\bar{\theta}_t^\lambda)_{t \geq 0}$ defined in (15), while all detailed proofs are deferred to Appendix A.7.

Lemma 3.9 (Moment bounds of TUSLA). *Let Assumptions 1, 2 and 3 hold. Then the following holds:*

(i) *For any $\lambda \in (0, \lambda_1)$, $n \in \mathbb{N}$, and $t \in (n, n+1]$,*

$$\mathbb{E} [|\bar{\theta}_t^\lambda|^2] \leq \left(1 - \frac{a^* \lambda(t-n)}{2}\right) \left(1 - \frac{\lambda a^*}{2}\right)^n \mathbb{E} [|\theta_0|^2] + c_0 \left(1 + \frac{2}{a^*}\right),$$

where $\lambda_1 := \min \left\{1, \frac{(a^)^2}{4K^4}\right\}$ and the constant c_0 is given by*

$$c_0 := \frac{a^*}{2} + 2b^* + 2d\beta^{-1} + 2K^2.$$

(ii) *For any $p \in [2, \infty) \cap \mathbb{N}$, $\lambda \in (0, \lambda_2)$, and $t \in (n, n+1]$,*

$$\mathbb{E} [|\bar{\theta}_t^\lambda|^{2p}] \leq \left(1 - \lambda(t-n) \frac{a^* \kappa(2)}{2}\right) \left(1 - \lambda \frac{a^* \kappa(2)}{2}\right)^n \mathbb{E} [|\theta_0|^{2p}] + c_3(p) \left(1 + \frac{2}{a^* \kappa(2)}\right),$$

where $\lambda_2 := \min \left\{1, \frac{(a^)^2}{4K^4}, \frac{1}{(a^*)^2}\right\}$ and the constants $c_3(p)$ and $\kappa(2)$ are given explicitly in (67) and (68).*

Lemma 3.10. *Let Assumptions 1, 2 and 3 hold. Then for any $0 < \lambda \leq \min \{1, (a^*)^2/(4K^4), 1/(a^*)^2\}$ and $t \in \mathbb{R}_+$, one obtains*

$$\mathbb{E} \left[\left| \bar{\theta}_t^\lambda - \bar{\theta}_{[t]}^\lambda \right|^4 \right] \leq \lambda^2 \left\{ 64K^4 \exp \left(-\frac{\lambda a^* \kappa(2)}{2} [t] \right) \mathbb{E} [|\theta_0|^{4r+4}] + \bar{C}_1 \right\},$$

where the constant $\bar{C}_1$ is given explicitly in (69).

Lemma 3.11. *Let Assumptions 1, 2 and 3 hold. Then for any $0 < \lambda \leq \min \{1, (a^*)^2/(4K^4), 1/(a^*)^2\}$, one obtains, for all $n \in \mathbb{N}$,*

$$\mathbb{E} \left[\left| h(\bar{\theta}_n^\lambda) - h_\lambda(\bar{\theta}_n^\lambda) \right|^2 \right] \leq C_1^\dagger \lambda$$

and for all $t \in [n, n+1)$, $n \in \mathbb{N}$,

$$\mathbb{E} \left[\left| h(\bar{\theta}_t^\lambda) - h(\bar{\theta}_n^\lambda) \right|^2 \right] \leq C_2^\dagger \lambda,$$

where the constants $C_1^\dagger, C_2^\dagger$ are given explicitly in (70) and (75).

3.4. Proof of Theorem 2.11. Building on the fundamental results outlined in Lemmas 7.1 to 7.3 of [35], particularly those pertaining to the interchangeability of integration and differentiation operations, we can rigorously establish a time-differential inequality governing the Kullback-Leibler divergence. This inequality aligns with the classical result presented in [55, Theorem 1]. Specifically, the derivation involves adapting the original formulation to accommodate our tamed numerical scheme, preserving the core analytical framework while incorporating adjustments to account for the modified discretization methodology.

Lemma 3.12 ([35]). *Let Assumptions 1, 2 and 3 hold. Then, for all $t \in [n, n+1)$,*

$$\mathbb{E} \left[\frac{\partial \pi_{\bar{\theta}_t^\lambda | \bar{\theta}_n^\lambda}(x | \bar{\theta}_n^\lambda)}{\partial t} \right] = \frac{\partial \pi_t(x)}{\partial t}, \quad \mathbb{E} \left[\Delta_x \pi_{\bar{\theta}_t^\lambda | \bar{\theta}_n^\lambda}(x | \bar{\theta}_n^\lambda) \right] = \Delta \pi_t(x)$$

and

$$\mathbb{E} \left[\nabla_x \cdot \left\{ \pi_{\bar{\theta}_t^\lambda | \bar{\theta}_n^\lambda}(x | \bar{\theta}_n^\lambda) h_\lambda(\bar{\theta}_n^\lambda) \right\} \right] = \nabla_x \cdot \left\{ \pi_t(x) \mathbb{E} \left[h_\lambda(\bar{\theta}_n^\lambda) | \bar{\theta}_t^\lambda = x \right] \right\}.$$

Proof. See Appendix A.8. $\square$

Corollary 3.13. *Let Assumptions 1, 2 and 3 hold. Then, for all $t \in [n, n+1)$,*

$$\partial_t \pi_t(x) = \lambda \left\{ \nabla \cdot \left(\pi_t(x) \mathbb{E} \left[h_\lambda(\bar{\theta}_n^\lambda) | \bar{\theta}_t^\lambda = x \right] \right) + \beta^{-1} \Delta \pi_t(x) \right\}.$$

Proof. Taking expectations in (16) and applying Lemma 3.12 yields the desired result. $\square$

Proposition 3.14. *Let Assumptions 1, 2 and 3 hold. Then for every $t \in [n, n+1]$ and $n \in \mathbb{N}$,*

$$\frac{d}{dt} \text{KL}(\pi_t | \pi_\beta) = -\lambda \int_{\mathbb{R}^d} \left\langle \pi_t(x) \mathbb{E} \left[h_\lambda(\bar{\theta}_n^\lambda) | \bar{\theta}_t^\lambda = x \right] + \frac{\nabla \pi_t(x)}{\beta}, \nabla \log \pi_t(x) - \nabla \log \pi_\beta(x) \right\rangle dx.$$

Proof. On the one hand, through straightforward calculation, one obtains, for all $t \in [n, n+1]$,

$$\frac{d}{dt} \text{KL}(\pi_t | \pi_\beta) = \int_{\mathbb{R}^d} \frac{\partial \pi_t(x)}{\partial t} (1 + \log \pi_t(x) - \log \pi_\beta(x)) dx. \quad (17)$$

Let

$$F_t(x) := \pi_t(x) \mathbb{E} \left[h_\lambda(\bar{\theta}_n^\lambda) | \bar{\theta}_t^\lambda = x \right] + \beta^{-1} \nabla \pi_t(x)$$

and

$$G_t(x) := 1 + \log \pi_t(x) - \log \pi_\beta(x).$$

Recall from Corollary 3.13 that

$$\frac{\partial \pi_t}{\partial t}(x) = \lambda \left\{ \nabla \cdot \left(\pi_t(x) \mathbb{E} \left[h_\lambda(\bar{\theta}_n^\lambda) | \bar{\theta}_t^\lambda = x \right] \right) + \beta^{-1} \Delta \pi_t(x) \right\} = \lambda (\nabla \cdot F_t)(x). \quad (18)$$

Since $\nabla \pi_t = \pi_t \nabla \log \pi_t$ using the Lemmas A.7, A.8 and 7.5 in [35], one deduces that there exists constants $C, q, r_\star > 0$ independent of $x \in \mathbb{R}^d$, uniform in a small neighborhood of $t > 0$, such that

$$\max \{ |F_t(x) G_t(x)|, |(\nabla \cdot F_t)(x) G_t(x)|, |\langle F_t(x), \nabla G_t(x) \rangle| \} \leq C(1 + |x|^q) e^{-r_\star |x|^2}. \quad (19)$$

For simplicity and convenience, we drop the dependence of the constants on t since we want to integrate with respect to $x \in \mathbb{R}^d$. Let $R^\star > 0$ and $n(x)$ be the outside normal unit vector on $\partial B(0, R^\star)$. Thus, by using (19), one gets

$$\int_{\partial B(0, R^\star)} \langle G_t(x) F_t(x), n(x) \rangle dx \leq C' (R^\star)^d (1 + |R^\star|^q) e^{-r_\star |x|^2}. \quad (20)$$

On the other hand, by applying the divergence theorem on $B(0, R^\star)$ and (20), one obtains

$$\int_{B(0, R^\star)} (\nabla F_t)(x) G_t(x) dx = - \int_{B(0, R^\star)} \langle F_t(x), \nabla G_t(x) \rangle dx + \int_{\partial B(0, R^\star)} \langle F_t(x) G_t(x), n(x) \rangle dx,$$

which further implies

$$\int_{\mathbb{R}^d} (\nabla \cdot F_t)(x) G_t(x) dx$$

$$\begin{aligned}
&= \lim_{R^* \rightarrow \infty} \int_{\partial B(0, R^*)} (\nabla \cdot F_t)(x) G_t(x) dx \\
&= \lim_{R^* \rightarrow \infty} \left(- \int_{B(0, R^*)} \langle F_t(x), \nabla G_t(x) \rangle dx + \int_{\partial B(0, R^*)} \langle F_t(x) G_t(x), n(x) \rangle dx \right) \\
&= - \lim_{R^* \rightarrow \infty} \int_{B(0, R^*)} \langle F_t(x), \nabla G_t(x) \rangle dx = - \int_{\mathbb{R}^d} \langle F_t(x), \nabla G_t(x) \rangle dx.
\end{aligned} \tag{21}$$

By substituting (18) and (21) into (17), one obtains

$$\frac{d}{dt} \text{KL}(\pi_t | \pi_\beta) = -\lambda \int_{\mathbb{R}^d} \langle F_t(x), \nabla G_t(x) \rangle dx = -\lambda \int_{\mathbb{R}^d} \langle F_t(x), \nabla \log \pi_t(x) - \nabla \log \pi_\beta(x) \rangle dx,$$

which completes the proof. $\square$

Theorem 3.15. *Suppose that Assumption 1, 2 and 3 are satisfied. Then for $0 < \lambda < \min \left\{ 1, \frac{(a^*)^2}{4K^4}, \frac{1}{(a^*)^2} \right\}$ and every $t \in [n, n+1]$, $n \in \mathbb{N}$, there holds*

$$\frac{d}{dt} \text{KL}(\pi_t | \pi_\beta) \leq - \left(\frac{3\lambda}{4\beta} \right) I_{\pi_\beta}(\pi_t) + \beta\lambda \mathbb{E} \left[\left| h(\bar{\theta}_t^\lambda) - h_\lambda(\bar{\theta}_n^\lambda) \right|^2 \right],$$

where $I_{\pi_\beta}(\pi_t)$ is the relative Fisher information of π_t with respect to π_β .

Proof. See Appendix A.8. $\square$

Proof of Theorem 2.11. By setting $\hat{c} := \left(\frac{3}{2\beta} \right) C_{\text{LSI}} > 0$ and using the differential inequality obtain in Theorem 3.15, one obtains for any $t \in [n, n+1]$,

$$\begin{aligned}
\frac{d}{dt} \text{KL}(\pi_t | \pi_\beta) &\leq - \left(\frac{3\lambda}{4\beta} \right) I_{\pi_\beta}(\pi_t) + \beta\lambda \mathbb{E} \left[\left| h(\bar{\theta}_t^\lambda) - h_\lambda(\bar{\theta}_n^\lambda) \right|^2 \right] \\
&\leq -(\hat{c}\lambda) \text{KL}(\pi_t | \pi_\beta) + 2\beta\lambda \mathbb{E} \left[\left| h(\bar{\theta}_t^\lambda) - h(\bar{\theta}_n^\lambda) \right|^2 \right] + 2\beta\lambda \mathbb{E} \left[\left| h(\bar{\theta}_n^\lambda) - h_\lambda(\bar{\theta}_n^\lambda) \right|^2 \right] \\
&\leq -(\hat{c}\lambda) \text{KL}(\pi_t | \pi_\beta) + 2\beta(C_1^\dagger + C_2^\dagger)\lambda^2,
\end{aligned}$$

where the first term has been bounded using logarithmic Sobolev inequality with constant $C_{\text{LSI}} > 0$ and the rest of the terms using the one-step error in Lemma 3.11. Splitting the terms one obtains

$$\left\{ \frac{d}{dt} \text{KL}(\pi_t | \pi_\beta) + (\hat{c}\lambda) \text{KL}(\pi_t | \pi_\beta) \right\} e^{\hat{c}\lambda t} \leq 2\lambda^2\beta(C_1^\dagger + C_2^\dagger) e^{\hat{c}\lambda t}.$$

Integrating over interval $[n, t]$ yields

$$e^{\hat{c}\lambda t} \text{KL}(\pi_t | \pi_\beta) - e^{n\lambda\hat{c}} \text{KL}(\pi_n | \pi_\beta) \leq \lambda \left\{ \frac{2\beta(C_1^\dagger + C_2^\dagger)}{\hat{c}} \right\} (e^{\hat{c}\lambda t} - e^{\hat{c}\lambda n}),$$

which leads to the following

$$\text{KL}(\pi_t | \pi_\beta) \leq e^{-\hat{c}(t-n)\lambda} \text{KL}(\pi_n | \pi_\beta) + \lambda \left\{ \frac{2\beta(C_1^\dagger + C_2^\dagger)}{\hat{c}} \right\} (1 - e^{-\hat{c}\lambda(t-n)}).$$

Furthermore, by taking $t = n+1$, one gets

$$\begin{aligned}
\text{KL}(\pi_{n+1} | \pi_\beta) &\leq e^{-\hat{c}\lambda} \text{KL}(\pi_n | \pi_\beta) + \lambda \left\{ \frac{2\beta(C_1^\dagger + C_2^\dagger)}{\hat{c}} \right\} (1 - e^{-\hat{c}\lambda}) \\
&\leq e^{-2\hat{c}\lambda} \text{KL}(\pi_{n-1} | \pi_\beta) + \lambda \left\{ \frac{2\beta(C_1^\dagger + C_2^\dagger)}{\hat{c}} \right\} (1 - e^{-\hat{c}\lambda}) (1 + e^{-\hat{c}\lambda}) \\
&\leq e^{-(n+1)\hat{c}\lambda} \text{KL}(\pi_0 | \pi_\beta) + \lambda \left\{ \frac{2\beta(C_1^\dagger + C_2^\dagger)}{\hat{c}} \right\},
\end{aligned}$$

which completes the proof. $\square$

4. NUMERICAL EXAMPLES

In this section, we seek to illustrate and verify our theoretical results with some numerical simulations. By running the Tamed Unadjusted Stochastic Langevin Algorithm, we extract samples from some high-dimensional target distributions, where the choices of target distribution $\pi_1(\theta)$ is proportional to $\theta \mapsto \exp(-U(\theta)) d\theta$.

4.1. Gaussian Mixture Model. In this subsection, we will show the performance of the proposed method in sampling the Gaussian mixture distribution. For fixed $\dot{m} \in \mathbb{R}^d$ and $|\dot{m}| > 1$, the density of d -dimensional Gaussian mixture example is defined as

$$\pi_1(\theta) = \frac{1}{2\sqrt{(2\pi)^d}} \left\{ \exp\left(-\frac{|\theta - \dot{m}|^2}{2}\right) + \exp\left(-\frac{|\theta + \dot{m}|^2}{2}\right) \right\}, \quad \forall \theta \in \mathbb{R}^d.$$

By simple calculation, the Gaussian mixture model has the form

$$\pi_1(\theta) \propto \exp(-U(\theta)) = e^{-|\theta - \dot{m}|^2/2} + e^{-|\theta + \dot{m}|^2/2}, \quad \dot{m} \in \mathbb{R}^d \text{ and } |\dot{m}| > 1,$$

where

$$U(\theta) = |\theta - \dot{m}|^2/2 - \log\left(1 + e^{-2\dot{m}^\top \theta}\right)$$

and U is the non-convex function on $\mathbb{R}^d$ (see Dalayan [15] for more details).

Proposition 4.1. *The gradient $h : \mathbb{R}^d \rightarrow \mathbb{R}^d$ of the TUSLA (4) is given by*

$$h(\theta) = \theta - \dot{m} + \frac{2\dot{m}}{1 + e^{2\dot{m}^\top \theta}}.$$

In addition, Assumptions 2 and 3 are satisfied with

$$L := 1 + 4|\dot{m}|^2, \quad K := \max\{1, |\dot{m}|\}, \quad \bar{a} := \frac{1}{2}, \quad \bar{b} := 2, \quad r := 0.$$

Proof. See Appendix A.9. □

4.2. Double-Well Model. For the second example we consider sampling from $\pi_1(\theta) \propto \exp(-U(\theta))$, where U is a double-well potential function given by

$$U(\theta) = \frac{1}{4}|\theta|^4 - \frac{1}{2}|\theta|^2, \quad \theta \in \mathbb{R}^d.$$

In practical applications, the double-well potential function is one of several quartic potentials of substantial importance in quantum mechanics and quantum field theory [27] that are of great significance for studying different physical phenomena or mathematical characteristics.

Proposition 4.2. *The gradient $h : \mathbb{R}^d \rightarrow \mathbb{R}^d$ of the TUSLA (4) is given by*

$$h(\theta) = \theta(|\theta|^2 - 1).$$

In addition, Assumptions 2 and 3 are satisfied with

$$L := 1, \quad K := 2, \quad r := 2, \quad a := 1/2, \quad b := 0, \quad \bar{r} := 0.$$

Proof. See Appendix A.9. □

4.3. Ginzburg-Landau Model. As a third example we consider the Ginzburg-Landau model of phase transitions in condensed-matter physics proposed by Ginzburg and Landau [24, Chapter 5], which describes a thermodynamic system undergoing continuous phase transitions, from a high-temperature symmetric phase to a low-temperature ordered phase in which some symmetry is broken.

In this subsection, we consider a three-dimensional 10^3 lattice, where each site represents a random variable $x = (x_{i,j,k})_{i,j,k \in \{1, \dots, 10\}} \in \mathbb{R}^{10^3}$. The corresponding potential is given for the 10^3 -dimensional parameter x by

$$U(x) := \sum_{i,j,k=1}^{10} \left\{ \frac{1-\tau}{2} x_{i,j,k}^2 + \frac{\alpha\tau}{2} |\tilde{\nabla} x_{i,j,k}|^2 + \frac{\zeta\tau}{4} x_{i,j,k}^4 \right\}, \quad (22)$$

where $\tau, \alpha, \zeta > 0$ and $\tilde{\nabla} x_{i,j,k} = (x_{i_+,j,k} - x_{i,j,k}, x_{i,j,k_+} - x_{i,j,k}, x_{i,j,k_+} - x_{i,j,k})$, where $i_+ = (i+1) \bmod 10$ and j_+, k_+ are defined similarly. The modular structure imposes periodic boundary conditions on the lattice. When the parameter $\tau < 1$, the conditional distribution for each $x_{i,j,k}$ looks Gaussian in

TABLE 1. Comparison of the first and second moments for each method (1D).

Moments	Model	Initial Value	TULA	MALA	RWM	TUSLA
$\mathbb{E}[X_1]$	G-M	-10	0.0865	2.9894	0.0089	0.0242
		0	0.1955	0.1269	0.1147	0.0334
		10	0.1986	2.5519	0.1682	0.2293
	D-W	0	0.0206	0.0457	0.0074	0.0040
		10	0.0125	0.0463	0.0066	0.0070
$\mathbb{E}[X_1 ^2]$	G-M	-10	0.0194	0.2777	0.0201	0.2706
		0	0.1244	0.0072	0.1754	0.0124
		10	0.0327	0.0130	0.2123	0.0844
	D-W	0	0.0445	0.0487	0.0285	0.1034
		10	0.0380	0.1271	0.0404	0.0922

TABLE 2. Comparison of the first and second moments for each method (2D).

Moments	Model	Initial Value	TULA	MALA	RWM	TUSLA
$\mathbb{E}[X_1]$	G-M	$(-10, 0)$	0.0494	2.0154	0.0428	0.0352
		$(0, 0)$	0.2377	0.0813	0.0964	0.3077
		$(10, 0)$	0.0148	1.9885	0.1514	0.1374
	D-W	$(0, 0)$	0.0171	0.0107	0.0055	0.0250
		$(10, 0)$	0.0173	0.0744	0.0045	0.0016
$\mathbb{E}[X_1 ^2]$	G-M	$(-10, 0)$	0.0652	0.0064	0.0499	0.0785
		$(0, 0)$	0.0086	0.0220	0.0088	0.0405
		$(10, 0)$	0.0064	0.0448	0.1165	0.1321
	D-W	$(0, 0)$	0.2109	0.2136	0.2217	0.1612
		$(10, 0)$	0.1991	0.1096	0.2079	0.1649

TABLE 3. Comparison of the first and second moments for each method (100D).

Moments	Model	Initial Value	TULA	MALA	RWM	TUSLA
$\mathbb{E}[X_1]$	G-M	$(-10, 0^{\otimes 99})$	0.0137	0.3302	0.9464	0.0360
		$(0, 0^{\otimes 99})$	0.0317	0.0338	0.0082	0.0072
		$(10, 0^{\otimes 99})$	0.0199	0.3708	0.9490	0.0205
	D-W	$(0, 0^{\otimes 99})$	0.0010	0.0018	0.0037	5.5464e-04
		$(1, 0^{\otimes 99})$	0.0029	0.0007	0.0167	3.5122e-05
$\mathbb{E}[X_1 ^2]$	G-M	$(-10, 0^{\otimes 99})$	0.9093	0.1402	3.3283	0.2265
		$(0, 0^{\otimes 99})$	0.9317	0.1501	0.1247	0.2476
		$(10, 0^{\otimes 99})$	0.8179	0.1962	3.3975	0.2532
	D-W	$(0, 0^{\otimes 99})$	0.1135	0.1053	0.1025	0.1069
		$(1, 0^{\otimes 99})$	0.1142	0.1045	0.1161	0.1019

the centre of the space but has a lighter tails. The model exhibits a phase transition at $\tau = 1$, so that when $\tau > 1$ each conditional distribution is bimodal. The parameter α controls the strength of correlation between neighbouring lattice sites, with larger values making the sampling problem more challenging. When $\tau < 1$, each $x_{i,j,k}$ is likely to be close to zero, while in the bimodal phase, they are more likely to be non-negligible in magnitude. When α is large, neighbouring parameters are likely to have the same sign. The bimodal phase therefore represents the system in its ordered state, whereas the system is disordered for $\tau < 1$. In addition, the core issue of statistical inference is how to sample distribution π_1 ,

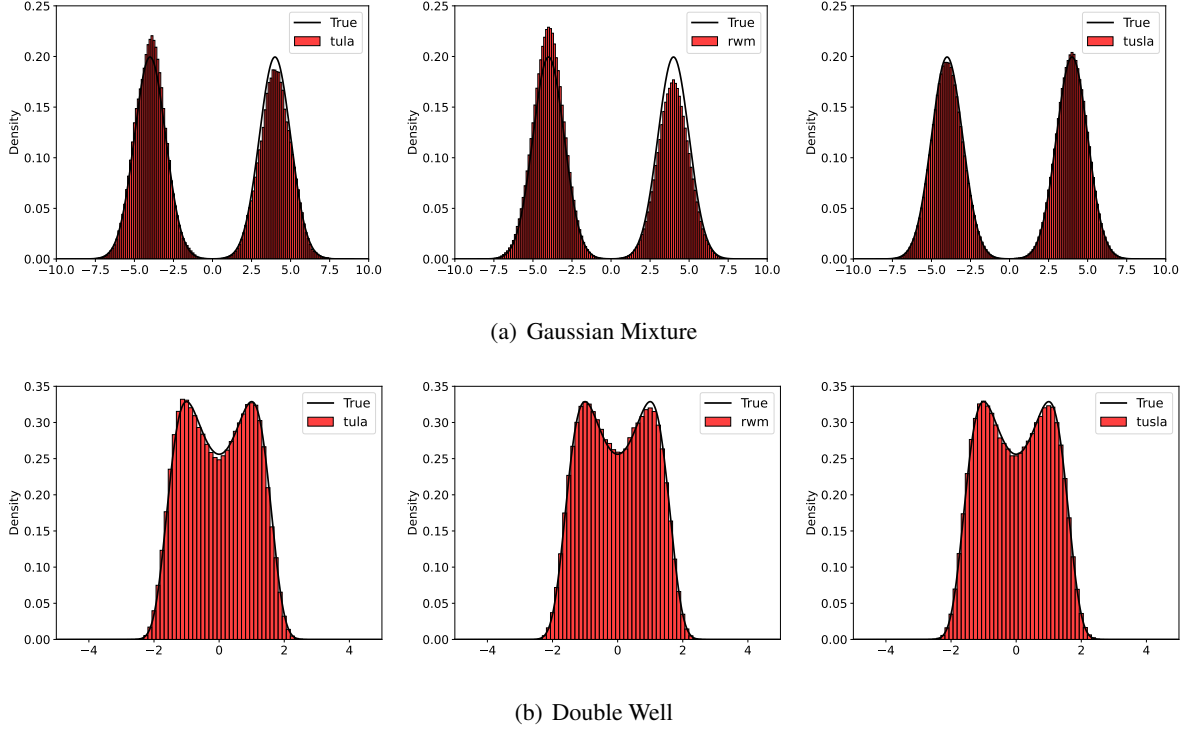


FIGURE 1. Comparison of TULA, RWM, and TUSLA for a Gaussian-Mixture and a Double-Well potential in one dimension ($\lambda = 0.001$).

TABLE 4. Comparison of the first and second moments for each method (1000D Ginzburg-Landau model).

Moments	Initial Value	TULA	MALA	RWM	TUSLA
$\mathbb{E}[X_1]$	$(10, 0^{\otimes 999})$	0.0238	0.0843	0.9017	0.0012
$\mathbb{E}[X_1 ^2]$	$(10, 0^{\otimes 999})$	0.2302	0.2396	0.8588	0.2192

which is defined by

$$\pi_1(dx) \propto \exp(-U(x)) dx, \quad x = (x_{i,j,k})_{i,j,k \in \{1,2,\dots,10\}} \in \mathbb{R}^{10^3}$$

where the Ginzburg-Landau model potential U is defined in (22).

Proposition 4.3. *For all $x = (x_{i,j,k})_{i,j,k \in \{1,\dots,d\}} \in \mathbb{R}^{d^3}$ with $d = 10$, the Ginzburg-Landau model gradient $h : \mathbb{R}^{d^3} \rightarrow \mathbb{R}^{d^3}$ of the TUSLA (4) is given by*

$$h(x) = \left\{ \alpha \tau (6x_{i,j,k} - x_{i+1,j,k} - x_{i,j+1,k} - x_{i,j,k+1} - x_{i-1,j,k} - x_{i,j-1,k} - x_{i,j,k-1}) \right. \\ \left. + (1 - \tau)x_{i,j,k} + \zeta \tau x_{i,j,k}^3 \right\}_{i,j,k \in \{1,2,\dots,d\}}.$$

In addition, Assumptions 1, 2 and 3 are satisfied with

$$L := \max \{ |1 - \tau| + 12\tau\alpha, 2\tau\zeta \}, \quad r := 2, \quad \bar{r} := 0, \\ K := \max \{ |1 - \tau| + 12\tau\alpha, \tau\zeta \}, \quad a := (\zeta\tau)/2, \quad b := |1 - \tau|.$$

Proof. See Appendix A.9. □

In our numerical experiments, we benchmark TUSLA against TULA given by [7], Metropolis Adjusted Langevin Algorithm (MALA), and Random Walk Metropolis algorithm (RWM). RWM is an MCMC algorithm using Gaussian proposal. That is, let $\{U_n\}_{n=1}^\infty$ and $\{\xi_n\}_{n=1}^\infty$ be i.i.d. sequences, with $U_n \sim \text{Unif}[0, 1]$ and $\xi_n \sim N(0, \mathbf{I}_d)$. Then the $(n + 1)$ -th RWM iterate writes a

$$\theta_{n+1}^\lambda = \theta_n^\lambda + \mathbb{1}_{\{U_{n+1} \leq \alpha(\theta_n^\lambda, \theta_n^\lambda + \sqrt{\lambda}\xi_{n+1})\}} (\sqrt{\lambda}\xi_{n+1}),$$

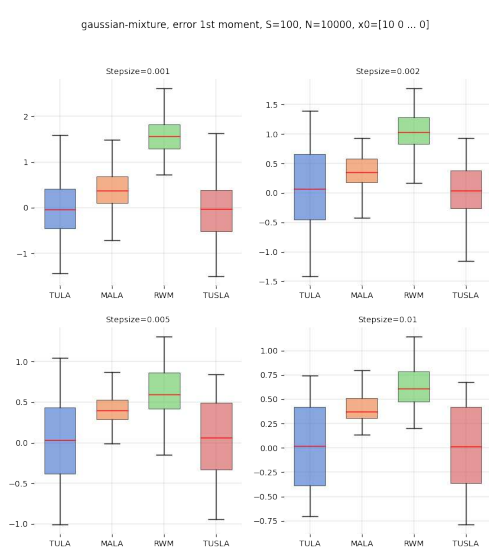


FIGURE 2. Boxplots of the error on the first moment for the Gaussian-Mixture (first coordinate) in dimension 100 starting at $(10, 0^{\otimes 99})$ for different step sizes.

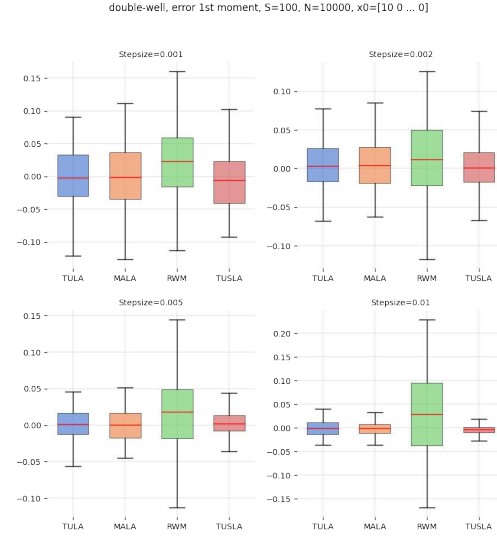


FIGURE 3. Boxplots of the error on the first moment for the double well (first coordinate) in dimension 100 starting at $(10, 0^{\otimes 99})$ for different step sizes.

where $\alpha(x, y) = \min\{1, \pi_1(y)/\pi_1(x)\}$ is the acceptance ratio. We set the dimension as $d \in \{1, 2, 100\}$ for the Gaussian Mixture and double well examples, and $d \in \{1, 1000\}$ for Ginzburg-Landau model.

On the one hand, we assume different initial values to study the convergence speed between different algorithms. On the other hand, it is numerically verified that the proposed algorithm is ergodic, i.e., the stationary distribution does not depend on the choice of the initial distribution. Therefore, the proposed algorithm can be used to sample the target distribution π_1 . For each algorithm, dimension, step size $\lambda \in \{0.001, 0.002, 0.005, 0.01\}$, we executed 100 independent Markov chains of 10^4 samples started at different initial conditions. For each run, we estimate the 1st and 2nd moments for the each coordinate, that is, $\int_{\mathbb{R}^d} \theta_i \pi_1(d\theta)$ and $\int_{\mathbb{R}^d} |\theta_i|^2 \pi_1(d\theta)$ for any $i \in \{1, \dots, d\}$. Since double well and Ginzburg-Landau models are coordinate-wise exchangeable, the results are provided only for their first coordinate.

For Gaussian Mixture, Double-Well and Ginzburg-Landau models, one obtains $\int_{\mathbb{R}^d} \theta_i \pi_1(d\theta) = 0$ for every $i \in \{1, \dots, d\}$. Moreover, by the symmetry and simple calculation, for the Gaussian Mixture and Double-Well models, we have for any $i \in \{1, \dots, d\}$, respectively,

$$\mathbb{E}[|X_i|^2] = \frac{1}{2} (2 + |\dot{m}_i|^2 + |-\dot{m}_i|^2) = 1 + |\dot{m}_i|^2$$

and for $r \in [0, +\infty)$,

$$\mathbb{E}[|X_i|^2] = \frac{1}{d} \left(\int_0^\infty r^2 v(r) dr \right) \left(\int_0^\infty v(r) dr \right)^{-1}, \quad v(r) := r^{d-1} \exp\left(-\frac{r^4}{4} + \frac{r^2}{2}\right).$$

The boxplots of empirical averages were computed to visually assess the performance distribution across different methods. For MALA and RWM, a stringent filtering criterion was applied: trajectories with acceptance ratios below 0.05 were excluded from the analysis to ensure only statistically meaningful samples were considered.

Selected boxplots are presented in Figures 2 to 4, which illustrate representative performance patterns across the tested scenarios. These visualizations consistently show that TUSLA maintains comparable result distributions to TULA, MALA, and RWM while exhibiting remarkable stability—characterized by narrower interquartile ranges and fewer outliers—across various experimental configurations.

Tables 1, 2 and 3 summarize the first and second moment errors for the benchmark Gaussian-Mixture (G-M) and Double-Well (D-W) models across different initializations in 1D, 2D and 100D settings,

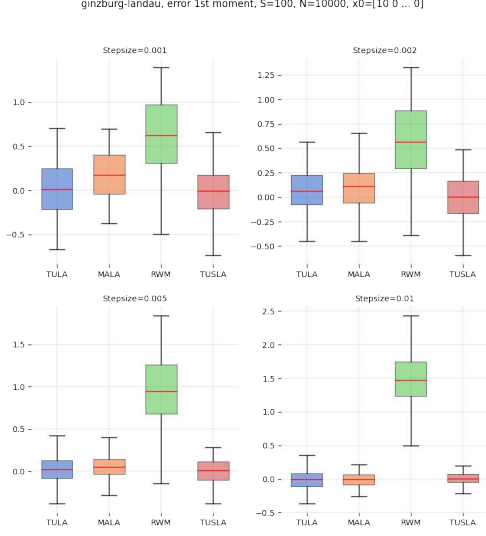


FIGURE 4. Boxplots of the error on the first moment for the Ginzburg-Landau model in dimension 1000 starting at $(10, 0^{\otimes 999})$ for different step sizes.

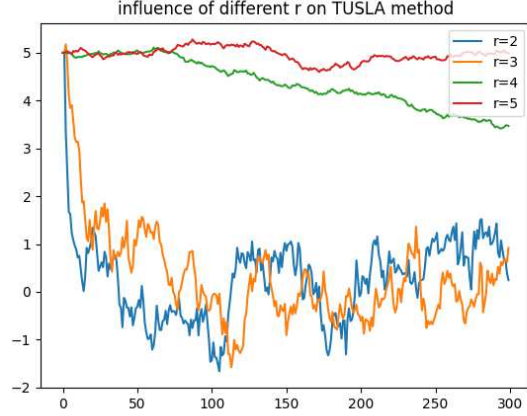


FIGURE 5. Effect of the parameter r on the convergence rate of TUSLA (1D Double-Well model).

respectively. Each table reports the estimation errors for $\mathbb{E}[X_1]$ and $\mathbb{E}[X_1^2]$ under various initial values (e.g., $-10, 0, 10$ in low dimensions; $(-10, 0^{\otimes 99})$, etc. in 100D). The results show that TUSLA matches or closely tracks the performance of TULA, MALA, and RWM in low and moderate dimensions for both Gaussian-Mixture and Double-Well models.

A striking advantage of TUSLA emerges in high-dimensional regimes. For example, in the 100D Double-Well model (Table 3), TUSLA achieves dramatically lower first-moment errors ($\mathbb{E}[X_1]$) than other methods, attaining $5.5464e-4$ and $3.5122e-5$ for two representative initializations—orders of magnitude below the corresponding errors of TULA, MALA, and RWM. This superiority also extends to the second moment estimation in 100D and is further corroborated in the 1000D Ginzburg-Landau experiment (see Table 4), where TUSLA consistently outperforms its competitors. These findings highlight that TUSLA is not only competitive in moderate-dimensional scenarios, but also exhibits robust and significant advantages for moment estimation in high-dimensional problems.

Additionally, we studied how the exponent r in $h_\lambda(\theta)$ (see (4)) affects TUSLA’s practical performance. As illustrated in Figure 5, setting $r = 2$ yields the best trade-off between convergence speed and estimation accuracy for the Double-Well model. We further observe that TUSLA’s convergence rate deteriorates as r increases, especially over longer optimization horizons, indicating that lower exponents are preferable in practice for fast and accurate moment estimation.

Figure 6 shows the trace plots of the first coordinate of the sample for the $d = 2$ dimensional Gaussian mixture model. As illustrated, TUSLA displays much faster stabilization and smaller oscillations compared to TULA and RWM, especially in the early stage of sampling. This demonstrates TUSLA’s advantage in rapidly reducing the influence of the initial value and achieving more stable sampling behavior. Such behavior is consistent with our observations in moment estimation (see Tables 3 and 4), further indicating the efficiency of TUSLA in high-dimensional settings.

Figure 7 compares the discrete total variation (TV) error for different sampling algorithms on the two-dimensional Double-Well model. The TV error is computed as the sum over the first two coordinates and averaged over 250,000 samples. As shown, TUSLA with $\lambda = 0.01$ and TULA both converge rapidly to a low TV error, demonstrating effective mixing and accurate approximation of the target distribution. In contrast, TUSLA with a larger value ($\lambda = 0.1$) displays a higher stationary TV error, indicating the sensitivity of performance to parameter choice. These results highlight the practical importance of tuning algorithm parameters and confirm the strong performance of TUSLA and TULA under appropriate settings.

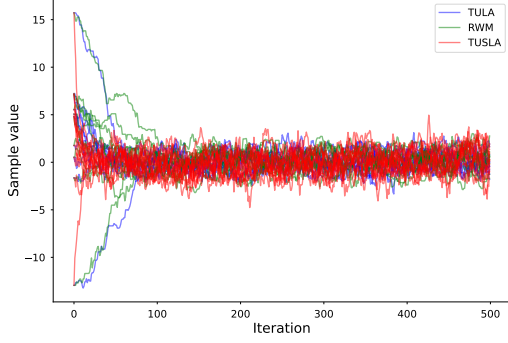


FIGURE 6. Trace plots of the first coordinate of the sample for the Gaussian mixture model in 2 dimensions.

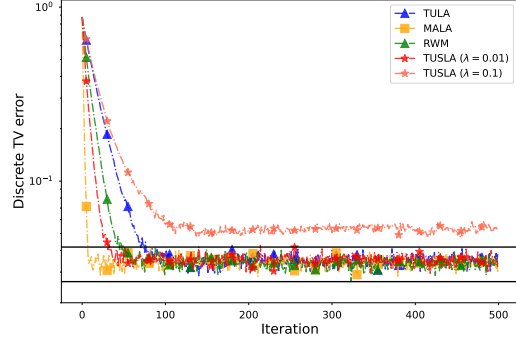


FIGURE 7. Discrete total variation (TV) error versus iteration for the Double-Well model ($d = 2$).

APPENDIX A. PROOF OF AUXILIARY RESULTS

A.1. Existence and uniqueness of solutions to Langevin diffusion.

Proof. For each $p \in [1, +\infty) \cap \mathbb{N}$, define the Lyapunov function $V \in C^2(\mathbb{R}^d; [1, +\infty))$ as follows

$$V(\theta) := 1 + |\theta|^{2p}, \quad \forall \theta \in \mathbb{R}^d.$$

By direct computation, one obtains

$$\nabla V(\theta) = 2p|\theta|^{2(p-1)}\theta$$

for all $\theta \in \mathbb{R}^d$ and

$$\begin{aligned} \Delta V(\theta) &= \text{tr} [\nabla^2 V(\theta)] = \text{tr} [2p|\theta|^{2(p-1)}\mathbf{I}_d + 2p(2p-2)|\theta|^{2(p-2)}\theta\theta^\top] \\ &= (2pd + 4p(p-1))|\theta|^{2(p-1)}. \end{aligned}$$

Therefore, using Remark 2.10 and the definition of $\mathcal{A}$ in (11) we obtain that for any $\theta \in \mathbb{R}^d$,

$$\begin{aligned} \mathcal{A}V(\theta) &= -\langle h(\theta), \nabla V(\theta) \rangle + \Delta V(\theta) \\ &\leq -2p|\theta|^{2(p-1)}(b^\sharp - a^*|\theta|^2) + (2pd + 4p(p-1))|\theta|^{2(p-1)} \\ &= 2pa^*|\theta|^{2p} - 2pb^\sharp|\theta|^{2(p-1)} + 2p(2p+d-2)|\theta|^{2(p-1)} \\ &\leq 1 + [2pa^* + 2p(2p+d-2)]|\theta|^{2p} \leq 2p(a^* + 2p+d-2)V(\theta). \end{aligned} \quad (23)$$

Hence, combining (23) and [29, Theorem 3.5], we obtain that there exists a unique strong solution of overdamped Langevin SDE (1) for any starting point $Z_0 = \theta_0$ with $\theta_0 \in \mathbb{R}^d$. $\square$

A.2. Proof of Remarks 2.10 and 2.12 in Section 2.

Proof of Remark 2.10. We first prove inequality (9) when $r > 0$. By assumption 3, one obtains, for any $\theta \in \mathbb{R}^d$,

$$\langle \theta, h(\theta) \rangle \geq a|\theta|^{r+2} - b|\theta|^{\bar{r}+2} + \langle \theta, h(0) \rangle \geq a|\theta|^{r+2} - b|\theta|^{\bar{r}+2} - |\theta| \cdot |h(0)|.$$

Therefore, by applying Cauchy-Schwarz's inequality and Assumption 2, we obtain

$$\begin{aligned} \langle \theta, h(\theta) \rangle &\geq a|\theta|^{r+2} - b|\theta|^{\bar{r}+2} - \frac{a}{2}|\theta|^2 - \frac{K^2}{2a} \\ &= \frac{a}{2}|\theta|^{r+2} + \left(\frac{a}{4}|\theta|^{r+2} - b|\theta|^{\bar{r}+2}\right) + \left(\frac{a}{4}|\theta|^{r+2} - \frac{a}{2}|\theta|^2\right) - \frac{K^2}{2a}. \end{aligned} \quad (24)$$

By using $0 \leq \bar{r} < r$, it follows that for any $\theta \in \mathbb{R}^d$,

$$\frac{a}{4}|\theta|^{r+2} - b|\theta|^{\bar{r}+2} > 0 \iff |\theta| > \left(\frac{4b}{a}\right)^{1/(r-\bar{r})}$$

and moreover,

$$\frac{a}{4}|\theta|^{r+2} - \frac{a}{2}|\theta|^2 > 0 \iff |\theta| > 2^{1/r}.$$

Denote by $D := \max \left\{ \left(\frac{4b}{a} \right)^{1/(r-\bar{r})}, 2^{1/r} \right\} > 1$. Then for any $|\theta| > D$, (24) becomes

$$\langle \theta, h(\theta) \rangle > \frac{a}{2}|\theta|^{r+2} - \frac{K^2}{2a}, \quad (25)$$

while for $|\theta| \leq D$, it follows that

$$\begin{aligned} \langle \theta, h(\theta) \rangle &\geq \frac{a}{2}|\theta|^{r+2} - b|\theta|^{\bar{r}+2} - \frac{a}{2}|\theta|^2 - \frac{K^2}{2a} \geq \frac{a}{2}|\theta|^{r+2} - bD^{\bar{r}+2} - \left(\frac{a}{2} \right) D^2 - \frac{K^2}{2a} \\ &\geq \frac{a}{2}|\theta|^{r+2} - \left(b + \frac{a}{2} \right) D^{\bar{r}+2} - \frac{K^2}{2a}. \end{aligned} \quad (26)$$

Combining (25) and (26) yields the inequality

$$\langle \theta, h(\theta) \rangle \geq \frac{a}{2}|\theta|^{r+2} - \left\{ \left(b + \frac{a}{2} \right) D^{\bar{r}+2} + \frac{K^2}{2a} \right\}. \quad (27)$$

Next, we consider the case $r = 0$, then Assumption 3 reads

$$\langle \theta, h(\theta) \rangle \geq \bar{a}|\theta|^2 - \bar{b} = \bar{a}|\theta|^{r+2} - \bar{b} \quad (28)$$

for all $\theta \in \mathbb{R}^d$. Therefore, combining (27) and (28), one obtains

$$\langle \theta, h(\theta) \rangle \geq a^*|\theta|^{r+2} - b^*,$$

for all $\theta \in \mathbb{R}^d$ and $r \in \mathbb{N}$, where

$$\begin{aligned} a^* &:= \frac{a}{2} \mathbb{1}_{\{r>0\}} + \bar{a} \mathbb{1}_{\{r=0\}}, \quad D := \max \left\{ \left(\frac{4b}{a} \right)^{1/(r-\bar{r})}, 2^{1/r} \right\}, \\ b^* &:= \left\{ \left(\frac{a}{2} + b \right) D^{2+\bar{r}} + \frac{K^2}{2a} \right\} \mathbb{1}_{\{r>0\}} + \bar{b} \mathbb{1}_{\{r=0\}}. \end{aligned}$$

Hence, this completes the proof of (9) of Remark 2.10.

Finally, we show the second inequality (10) of Remark 2.10, set

$$b^\sharp := (b^* + a^*) \mathbb{1}_{\{r>0\}} + \bar{b} \mathbb{1}_{\{r=0\}}.$$

When $r > 0$, it suffices to prove that for any $\theta \in \mathbb{R}^d$,

$$a^*|\theta|^{2+r} - b^* \geq a^*|\theta|^2 - (b^* + a^*),$$

which implies that

$$a^* (|\theta|^r - 1) |\theta|^2 + a^* \geq 0.$$

Hence, for all $|\theta| \geq 1$, then

$$a^*|\theta|^{r+2} - a^*|\theta|^2 + a^* \geq a^* \geq 0 \quad (29)$$

and for all $|\theta| \leq 1$, then

$$a^*|\theta|^{r+2} - a^*|\theta|^2 + a^* \geq -a^* + a^* = 0. \quad (30)$$

The veracity of the second inequality (10) of Remark 2.10 is established from (29) and (30) in the case $r > 0$. On the other hand, when $r = 0$, the second inequality of Remark 2.10 holds trivially. Therefore, this completes the proof. $\square$

Proof of Remark 2.12. By applying Assumption 2 and direct calculation, one obtains

$$\begin{aligned} U(\theta) &= \int_0^1 \langle h(t\theta), \theta \rangle dt + U(0) \\ &\leq \int_0^1 |h(t\theta)| \cdot |\theta| dt + U(0) \\ &\leq K \int_0^1 (|\theta| + t^{1+r} |\theta|^{2+r}) dt + U(0) \\ &= K \left(|\theta| + \frac{1}{2+r} |\theta|^{2+r} \right) + U(0). \end{aligned}$$

On the one hand, by the Jensen inequality, one gets, for any $x \in \mathbb{R}^d$,

$$|x|^p = \left(\sum_{i=1}^d |x_i|^2 \right)^{\frac{p}{2}} = d^{\frac{p}{2}} \left(\frac{1}{d} \sum_{i=1}^d |x_i|^2 \right)^{\frac{p}{2}} \leq d^{\frac{p-2}{2}} \left(\sum_{i=1}^d |x_i|^p \right),$$

which implies

$$\begin{aligned} \int_{\mathbb{R}^d} |x|^p \pi_0(dx) &\leq (2\pi)^{-\frac{d}{2}} \int_{\mathbb{R}^d} d^{\frac{p-2}{2}} \left(\sum_{i=1}^d |x_i|^p \right) \exp\left(-\frac{|x|^2}{2}\right) dx \\ &\leq \frac{d^{\frac{p-2}{2}}}{(2\pi)^{\frac{d}{2}}} \sum_{i=1}^d \left\{ \int_{\mathbb{R}^d} |x_i|^p \exp\left(-\frac{|x|^2}{2}\right) dx \right\} \\ &\leq \frac{d^{\frac{p-2}{2}}}{\sqrt{2\pi}} \sum_{i=1}^d \left\{ \int_{\mathbb{R}} |x_i|^p \exp\left(-\frac{|x_i|^2}{2}\right) dx_i \right\} \\ &\leq \frac{d^{\frac{p}{2}}}{\sqrt{2\pi}} \int_{\mathbb{R}} |x|^p \exp\left(-\frac{|x|^2}{2}\right) dx \leq \frac{(2d)^{\frac{p}{2}}}{\sqrt{2\pi}} \Gamma\left(\frac{1+p}{2}\right), \end{aligned} \quad (31)$$

where the last inequality follows $\mathbb{E}[|\theta|^p] = 2^{p/2} \Gamma\left(\frac{1+p}{2}\right) / \sqrt{\pi}$ for standard one-dimensional Gaussian random variable θ . On the other hand, by the definition of Kullback-Leibler divergence, one obtains

$$\begin{aligned} \text{KL}(\pi_0 \mid \pi_\beta) &= \int_{\mathbb{R}^d} \log\left(\frac{\pi_0(x)}{\pi_\beta(x)}\right) \pi_0(dx) \\ &= \int_{\mathbb{R}^d} \log(\pi_0(x)) \pi_0(dx) + \beta \int_{\mathbb{R}^d} U(x) \pi_0(dx) \\ &\leq \frac{d}{2} \log\left(\frac{1}{2\pi e}\right) + \beta \int_{\mathbb{R}^d} \left\{ U(0) + K \left(|x| + \frac{|x|^{2+r}}{2+r} \right) \right\} \pi_0(dx). \end{aligned}$$

Combining (31) and the following inequality

$$|x| = \left(\sum_{i=1}^d |x_i|^2 \right)^{\frac{1}{2}} \leq \left(\sum_{i=1}^d |x_i| \right)^{2 \cdot \frac{1}{2}} = \sum_{i=1}^d |x_i|,$$

one obtains the following

$$\begin{aligned} \text{KL}(\pi_0 \mid \pi_\beta) &\leq \frac{d}{2} \log\left(\frac{1}{2\pi e}\right) + \beta \left\{ U(0) + \frac{K}{2+r} \cdot \frac{(2d)^{\frac{2+r}{2}}}{\sqrt{\pi}} \Gamma\left(\frac{3+r}{2}\right) + K \int_{\mathbb{R}^d} \sum_{i=1}^d |x_i| \pi_0(dx) \right\} \\ &= \frac{d}{2} \log\left(\frac{1}{2\pi e}\right) + \beta \left\{ U(0) + \frac{K}{2+r} \cdot \frac{(2d)^{\frac{2+r}{2}}}{\sqrt{\pi}} \Gamma\left(\frac{3+r}{2}\right) + Kd \int_{\mathbb{R}} |x| \pi_0(dx) \right\} \\ &= \frac{d}{2} \log\left(\frac{1}{2\pi e}\right) + \beta \left\{ U(0) + \frac{K(2d)^{\frac{2+r}{2}}}{\sqrt{\pi}(2+r)} \Gamma\left(\frac{3+r}{2}\right) + Kd\sqrt{\frac{2}{\pi}} \right\}, \end{aligned}$$

which completes the proof. $\square$

A.3. Proof of Proposition 2.15 in Section 2.

Lemma A.1. Suppose that Assumptions 2-3 hold. Then, for any $\lambda \in \left(0, \frac{(a^*)^2}{4K^4}\right)$, the following holds

$$\liminf_{|\theta| \rightarrow +\infty} \left\{ \left\langle \frac{\theta}{|\theta|}, h_\lambda(\theta) \right\rangle - \frac{\lambda}{2|\theta|} |h_\lambda(\theta)|^2 \right\} > 0.$$

Proof. Indeed, by using the Assumption 2 and (9) in Remark 2.10, one obtains

$$\langle \theta, h_\lambda(\theta) \rangle - \frac{\lambda}{2} |h_\lambda(\theta)|^2 = \frac{\langle \theta, h(\theta) \rangle}{1 + \sqrt{\lambda}|\theta|^r} - \frac{\lambda |h(\theta)|^2}{2(1 + \sqrt{\lambda}|\theta|^r)^2} \geq \frac{a^*|\theta|^{r+2} - b^*}{1 + \sqrt{\lambda}|\theta|^r} - \frac{\lambda K^2(1 + |\theta|^{r+1})^2}{2(1 + \sqrt{\lambda}|\theta|^r)^2}.$$

By applying $(a + b)^2 \leq 2(a^2 + b^2)$ for any $a, b \geq 0$, then the above inequality implies

$$\langle \theta, h_\lambda(\theta) \rangle - \frac{\lambda}{2} |h_\lambda(\theta)|^2 \geq \frac{a^* |\theta|^{r+2} - b^*}{1 + \sqrt{\lambda} |\theta|^r} - (\lambda K^2) \frac{1 + |\theta|^{2(r+1)}}{(1 + \sqrt{\lambda} |\theta|^r)^2}. \quad (32)$$

For any $\theta \in \mathbb{R}^d$, we denote

$$\mathcal{I}(\theta) := \frac{b^*}{1 + \sqrt{\lambda} |\theta|^r} + \frac{\lambda K^2}{(1 + \sqrt{\lambda} |\theta|^r)^2}, \quad \mathcal{J}(\theta) := \frac{|\theta|^r}{1 + \sqrt{\lambda} |\theta|^r}.$$

So, by (32), one writes

$$\begin{aligned} \langle \theta, h_\lambda(\theta) \rangle - \frac{\lambda}{2} |h_\lambda(\theta)|^2 &\geq a^* |\theta|^2 \mathcal{J}(\theta) - (\lambda K^2) \frac{|\theta|^{2(r+1)}}{(1 + \sqrt{\lambda} |\theta|^r)^2} - \mathcal{I}(\theta) \\ &= \frac{a^*}{2} |\theta|^2 \mathcal{J}(\theta) + \left[\frac{a^*}{2} |\theta|^2 \mathcal{J}(\theta) - (\lambda K^2) \frac{|\theta|^{2(r+1)}}{(1 + \sqrt{\lambda} |\theta|^r)^2} \right] - \mathcal{I}(\theta). \end{aligned}$$

On the one hand, for any $\lambda > 0$, the function $f(t) := t/(1 + t\sqrt{\lambda})$ is non-decreasing for all $t \geq 0$. Thus, using $\lambda \in (0, 1)$, then one obtain

$$\mathcal{J}(\theta) \geq \frac{1}{1 + \sqrt{\lambda}} > \frac{1}{2}, \quad \text{for all } |\theta| \geq 1.$$

On the other hand, by the fact that $0 < \lambda \leq (a^*)^2/(4K^4)$, it yields that

$$\begin{aligned} \frac{a^*}{2} |\theta|^2 \mathcal{J}(\theta) - (\lambda K^2) \frac{|\theta|^{2(r+1)}}{(1 + \sqrt{\lambda} |\theta|^r)^2} &= \frac{(a^*/2) |\theta|^{r+2} (1 + \sqrt{\lambda} |\theta|^r) - (\lambda K^2) |\theta|^{2(r+1)}}{(1 + \sqrt{\lambda} |\theta|^r)^2}, \\ &\geq \frac{(a^*/2) \sqrt{\lambda} |\theta|^{2(r+1)} - (\lambda K^2) |\theta|^{2(r+1)}}{(1 + \sqrt{\lambda} |\theta|^r)^2}, \\ &\geq \frac{\sqrt{\lambda} \left(\frac{a^*}{2} - \sqrt{\frac{(a^*)^2}{4K^4}} K^2 \right) |\theta|^{2(1+r)}}{(1 + \sqrt{\lambda} |\theta|^r)^2} = 0. \end{aligned}$$

Therefore, through the above inequality, for any $\theta \in \mathbb{R}^d$ one obtains

$$\langle \theta, h_\lambda(\theta) \rangle - \frac{\lambda}{2} |h_\lambda(\theta)|^2 \geq \frac{a^*}{4} |\theta|^2 - \mathcal{I}(\theta), \quad (33)$$

and

$$\liminf_{|\theta| \rightarrow \infty} \frac{\mathcal{I}(\theta)}{|\theta|} = \liminf_{|\theta| \rightarrow \infty} \frac{1}{|\theta|} \left\{ \frac{b^*}{1 + \sqrt{\lambda} |\theta|^r} + \frac{\lambda K^2}{(1 + \sqrt{\lambda} |\theta|^r)^2} \right\} = 0. \quad (34)$$

Finally, combining (33) and (34), we get the desired estimate (A.1). $\square$

A.4. Proof of One-sided Lipschitz Condition in Section 2.

Lemma A.2. Under Assumptions 2 and 3 with $r > 0$, then for all $\theta, \theta' \in \mathbb{R}^d$, the inequality

$$\langle \theta - \theta', h(\theta) - h(\theta') \rangle \geq -\bar{L} |\theta - \theta'|^2$$

holds, where constants $\bar{L} := L(1 + 2\bar{R})^r$ and $\bar{R} := (b/a)^{1/(r-\bar{r})}$. Moreover, there exist constants $m > 0, R_\star > 0$ such that for any $\theta, \theta' \in \mathbb{R}^d$

$$\langle \theta - \theta', h(\theta) - h(\theta') \rangle \geq m |\theta - \theta'|^2$$

when $|\theta - \theta'| \geq R_\star$ holds.

Proof of Lemma A.2. By using Assumption 3 with $r > 0$, one obtains, for any $\theta, \theta' \in \mathbb{R}^d$,

$$\langle \theta - \theta', h(\theta) - h(\theta') \rangle \geq a(|\theta|^r + |\theta'|^r) |\theta - \theta'|^2 - b(|\theta|^{\bar{r}} + |\theta'|^{\bar{r}}) |\theta - \theta'|^2.$$

Moreover, observe that for all $\theta, \theta' \in \mathbb{R}^d$,

$$a|\theta - \theta'|^2 |\theta|^r - b|\theta - \theta'|^2 |\theta'|^{\bar{r}} > 0 \iff |\theta| > \left(\frac{b}{a} \right)^{1/(r-\bar{r})}. \quad (35)$$

Define $\bar{R} := \left(\frac{b}{a}\right)^{1/(r-\bar{r})}$, and denote by $\bar{B}(0, \bar{R})$ the closed ball with radius $\bar{R}$ centered on the zero vector in $\mathbb{R}^d$. Then, for $\theta, \theta' \notin \bar{B}(0, \bar{R})$, one obtains, by using (35),

$$\langle \theta - \theta', h(\theta) - h(\theta') \rangle > 0. \quad (36)$$

Furthermore, from the Cauchy-Schwarz inequality and Assumption 2, for all $\theta, \theta' \in \bar{B}(0, \bar{R})$,

$$\begin{aligned} -\langle \theta - \theta', h(\theta) - h(\theta') \rangle &\leq |\theta - \theta'| \cdot |h(\theta) - h(\theta')| \\ &\leq L(1 + |\theta| + |\theta'|)^r |\theta - \theta'|^2 \\ &\leq L(1 + 2\bar{R})^r |\theta - \theta'|^2. \end{aligned} \quad (37)$$

For the case $|\theta| \leq \bar{R}$ and $|\theta'| > \bar{R}$, it is straightforward to see that the inequality (36) holds when $|\theta| = \bar{R}$ and $|\theta'| > \bar{R}$. Moreover, for $|\theta| < \bar{R}$, $|\theta'| > \bar{R}$, consider $\bar{\theta} := \theta' + c_{\theta, \theta'}(\theta - \theta')$, where $c_{\theta, \theta'}$ is chosen as the unique scalar constant in the interval $(0, 1)$ such that $|\bar{\theta}| = \bar{R}$. Observe that $\theta - \bar{\theta} = (1 - c_{\theta, \theta'})(\theta - \theta')$. One then gets, for $|\theta| < \bar{R}$, $|\theta'| > \bar{R}$,

$$\begin{aligned} \langle \theta - \theta', h(\theta) - h(\theta') \rangle &= \langle \theta - \theta', h(\theta) - h(\bar{\theta}) \rangle + \langle \theta - \theta', h(\bar{\theta}) - h(\theta') \rangle \\ &= \langle \theta - \theta', h(\theta) - h(\bar{\theta}) \rangle + \frac{1}{c_{\theta, \theta'}} \langle \bar{\theta} - \theta', h(\bar{\theta}) - h(\theta') \rangle, \end{aligned} \quad (38)$$

where the equality above is obtained by using $\theta - \theta' = \frac{1}{c_{\theta, \theta'}}(\bar{\theta} - \theta')$. One notes the second term on the RHS of (38) is greater zero due to (36). Thus, it follows

$$\begin{aligned} \langle \theta - \theta', h(\theta) - h(\theta') \rangle &> \langle \theta - \theta', h(\theta) - h(\bar{\theta}) \rangle \\ &= \frac{1}{1 - c_{\theta, \theta'}} \langle \theta - \bar{\theta}, h(\theta) - h(\bar{\theta}) \rangle \\ &\geq -\frac{L(1 + 2\bar{R})^r}{1 - c_{\theta, \theta'}} |\theta - \bar{\theta}|^2 \end{aligned} \quad (39)$$

$$\begin{aligned} &= -L(1 + 2\bar{R})^r (1 - c_{\theta, \theta'}) |\theta - \theta'|^2 \\ &\geq -L(1 + 2\bar{R})^r |\theta - \theta'|^2, \end{aligned} \quad (40)$$

where the inequality (39) holds due to (37), and the last inequality follows from $c_{\theta, \theta'} \in (0, 1)$. Interchanging the role of θ and θ' in the above argument shows that (40) holds for $|\theta| > \bar{R}$ and $|\theta'| < \bar{R}$. Therefore, by denoting $\bar{L} := L(1 + 2\bar{R})^r$ and $\bar{R} := (b/a)^{1/(r-\bar{r})}$, one concludes

$$\langle \theta - \theta', h(\theta) - h(\theta') \rangle \geq -\bar{L} |\theta - \theta'|^2$$

for all $\theta, \theta' \in \mathbb{R}^d$. The proof of the results in the second part is very standard. For details, see [35, lemma A.11]. Finally, this completes the proof. $\square$

A.5. Proof of Proposition 3.2 and 3.5 in Section 3.

Proof of Proposition 3.2. By applying Proposition 3.1, one establishes

$$\frac{\beta}{C_{\text{LSI}}} \leq \frac{2(\bar{a})^2 + 2L^2}{(\bar{a})^2 L} + \frac{\beta}{\lambda_\star} \left\{ \frac{6L(d + \beta)}{\bar{a}} + 2 \right\} \quad (41)$$

and

$$\frac{\beta}{\lambda_\star} \leq \frac{1}{\bar{a}(d + \bar{b}\beta)} + \frac{2C_\star(d + \bar{b}\beta)}{\bar{a}} \exp \left\{ \frac{2}{\bar{a}} (L + B_\star)(\bar{b}\beta + d) + \beta(A_\star + B_\star) \right\}.$$

For the spectral gap term, it yields the following

$$\begin{aligned} \frac{\beta}{\lambda_\star} &\leq \frac{1}{\bar{a}d} + \frac{2C_\star(d + \bar{b}\beta)}{\bar{a}} e^{\frac{2d}{\bar{a}}(L+B_\star)} \cdot e^{(2(L+B_\star)(\bar{a})^{-1} + (A_\star+B_\star))\beta} \\ &\leq \frac{1}{\bar{a}d} + \frac{2C_\star d}{\bar{a}} e^{2d(L+B_\star)(\bar{a})^{-1}} \cdot e^{(2\bar{b}(L+B_\star)(\bar{a})^{-1} + (A_\star+B_\star+b))\beta} \\ &\leq \left(\frac{1}{\bar{a}d} + \frac{2C_\star d}{\bar{a}} e^{2d(L+B_\star)(\bar{a})^{-1}} \right) e^{(2\bar{b}(L+B_\star)(\bar{a})^{-1} + (A_\star+B_\star+b))\beta}, \end{aligned} \quad (42)$$

where the second inequality follows from $d + \bar{b}\beta \leq de^{\bar{b}\beta}$ for all $\beta > 0, d \geq 1$. Combining the inequalities (41) and (42) yields

$$\begin{aligned} \frac{\beta}{C_{\text{LSI}}} &\leq \frac{2(\bar{a})^2 + 2L^2}{(\bar{a})^2 L} + \frac{\beta}{\lambda_\star} \left\{ \frac{6L(d + \beta)}{\bar{a}} + 2 \right\} \\ &\leq \frac{2(\bar{a})^2 + 2L^2}{(\bar{a})^2 L} e^{(2\bar{b}(L+B_\star)(\bar{a})^{-1} + (A_\star + B_\star + \bar{b} + 1))\beta} + \frac{\beta}{\lambda_\star} \left\{ \frac{6Ld}{\bar{a}} + 2 \right\} e^\beta \\ &\leq \left\{ \frac{2(\bar{a})^2 + 2L^2}{(\bar{a})^2 L} + \left(2 + \frac{6Ld}{\bar{a}} \right) \left(\frac{1}{\bar{a}d} + \frac{2C_\star d}{\bar{a}} e^{2d(L+B_\star)(\bar{a})^{-1}} \right) \right\} \exp(\beta C_2^\#), \end{aligned}$$

where

$$C_2^\# := \frac{2\bar{b}}{\bar{a}}(L + B_\star) + (A_\star + B_\star + \bar{b} + 1).$$

Therefore, taking into account that a lower bound of a Log-Sobolev constant automatically satisfies the Log-Sobolev inequality, we establish the desired result. $\square$

Proof of Proposition 3.5. By combining the equivalence between Wasserstein distance and gradient contraction established in [30] with Proposition 3.4, one obtains

$$W_1(\mathcal{L}(Z_t), \mathcal{L}(Z'_t)) \leq \hat{c}e^{-\hat{c}t} W_1(\mathcal{L}(\theta_0), \mathcal{L}(\theta'_0)),$$

which implies the following gradient estimate

$$|\nabla P_t f|(x) \leq \hat{c} \exp(-\hat{c}t) P_t |\nabla f|(x), \quad (43)$$

where $(P_t)_{t \geq 0}$ be the semigroup associated with the Langevin SDE (1) and $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is any smooth function. By the Birkhoff ergodic theorem, as $t \rightarrow \infty$, $P_t f(x)$ converges to $\pi_\beta(f)$ for all smooth functions f on $\mathbb{R}^d$. Thus, for any smooth function f and $t \geq 0$,

$$\begin{aligned} P_t f^2(x) - (P_t f(x))^2 &= \int_0^t \partial_s (P_s (P_{t-s} f)^2(x)) \, ds \\ &= \int_0^t \{ P_s \mathcal{A} (P_{t-s} f)^2(x) - 2P_s (P_{t-s} f \mathcal{A} P_{t-s} f)(x) \} \, ds \\ &= 2\beta^{-1} \int_0^t P_s |\nabla P_{t-s} f|^2(x) \, ds \\ &\leq \left(2\beta^{-1} (\hat{c})^2 \int_0^t e^{-2\hat{c}s} \, ds \right) P_t |\nabla f|^2(x), \end{aligned}$$

where the last inequality follows from (43) and $(P_t |\nabla f|(x))^2 \leq P_t |\nabla f|^2(x)$. Finally, letting $t \rightarrow \infty$, one obtains

$$\text{Var}_{\pi_\beta}(f) = \int_{\mathbb{R}^d} f^2 \, d\pi_\beta - \left(\int_{\mathbb{R}^d} f \, d\pi_\beta \right)^2 \leq \left(\frac{\hat{c}}{\sqrt{\hat{c}\beta}} \right)^2 \int_{\mathbb{R}^d} |\nabla f|^2 \, d\pi_\beta,$$

which completes the proof. $\square$

A.6. Proof of Proposition 3.7 in Section 3.

Lemma A.3. *Under the assumptions of Lemma A.2, the following inequality holds for any $\delta \in (0, m/4)$,*

$$\int_{\mathbb{R}^{2d}} e^{\delta|x-y|^2} \pi_\beta \otimes \pi_\beta(dx, dy) \leq (1 + 4d + (2\bar{L} + 8\delta)R_\star^2) \exp \left(\delta \max \left\{ \frac{1 + 4d}{2(m - 4\delta)}, R_\star^2 \right\} \right).$$

Proof. Let $\mathcal{A}^{\otimes 2}$ denote the generator of two independent Langevin SDEs satisfying (1), so that $\pi_\beta \otimes \pi_\beta$ is invariant under $\mathcal{A}^{\otimes 2}$. For $\delta \in (0, m/4)$, consider the function $V(x, y) = e^{\delta|x-y|^2}$. A direct calculation yields

$$\frac{\mathcal{A}^{\otimes 2} V(x, y)}{V(x, y)} = -2\delta \langle x - y, h(x) - h(y) \rangle + 4\beta^{-1}\delta d + 8\beta^{-1}\delta^2|x - y|^2$$

Using the one-sided Lipschitz continuity condition from Lemma A.2, one can bound this expression in two regimes

$$\begin{aligned} \frac{\mathcal{A}^{\otimes 2}V(x, y)}{V(x, y)} &\leq \begin{cases} 4\delta d + (8\delta^2 - 2m\delta)|x - y|^2 & \text{if } |x - y| \geq R_* \\ 4\delta d + (2\bar{L}\delta + 8\delta^2)R_*^2 & \text{otherwise} \end{cases} \\ &\leq -\delta \mathbb{1}_{|x-y| \geq M} + \delta(4d + (2\bar{L} + 8\delta)R_*^2) \mathbb{1}_{|x-y| < M}, \end{aligned}$$

where the first inequality holds due to inverse temperature $\beta \geq 1$. In addition, let $M^2 := \max\left(\frac{1+4d}{2(m-4\delta)}, R_*^2\right)$, then one obtains

$$\begin{aligned} \mathcal{A}^{\otimes 2}V(x, y) &\leq -\delta V(x, y) + \delta V(x, y)(1 + 4d + 2(\bar{L} + 4\delta)R_*^2) \mathbb{1}_{|x-y| \leq M} \\ &\leq -\delta V(x, y) + \delta(1 + 4d + 2(\bar{L} + 4\delta)R_*^2)e^{\delta M^2}. \end{aligned}$$

Integrating with respect to $\pi_\beta \otimes \pi_\beta$ and noting that $\int_{\mathbb{R}^d \times \mathbb{R}^d} \mathcal{A}^{\otimes 2}V(x, y) \pi_\beta \otimes \pi_\beta(dx, dy) = 0$ by invariance, one gets

$$\int_{\mathbb{R}^d \otimes \mathbb{R}^d} V(x, y) \pi_\beta \otimes \pi_\beta(dx, dy) \leq (1 + 4d + 2R_*^2(\bar{L} + 4\delta))e^{\delta M^2},$$

which completes the proof. $\square$

Proof of Proposition 3.7. Let $f \geq 0$ be such that $\pi_\beta(f^\alpha) = 1$. By applying Proposition 3.6 with $K = \bar{L}R_*$, one gets for any $y \in \mathbb{R}^d$,

$$\begin{aligned} 1 &= \int_{\mathbb{R}^d} P_t(f^\alpha)(x) \pi_\beta(dx) \\ &\geq (P_t f(y))^\alpha \int_{\mathbb{R}^d} \exp\left(\frac{-\alpha}{4(\alpha-1)/\beta} \left(K^2 t + \frac{|x-y|^2}{t}\right)\right) \pi_\beta(dx). \end{aligned}$$

On the other hand, for any $\gamma > \alpha > 1$, one derives the following estimate

$$\begin{aligned} \int_{\mathbb{R}^d} (P_t f(y))^\gamma \pi_\beta(dy) &\leq \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \exp\left(\frac{-\alpha}{4(\alpha-1)/\beta} \left(K^2 t + \frac{|x-y|^2}{t}\right)\right) \pi_\beta(dx) \right]^{-\gamma/\alpha} \pi_\beta(dy) \\ &\leq \int_{\mathbb{R}^d} \left\{ \int_{\mathbb{R}^d} \exp\left(\frac{\gamma}{4(\alpha-1)/\beta} \left(K^2 t + \frac{|x-y|^2}{t}\right)\right) \pi_\beta(dx) \right\} \pi_\beta(dy). \end{aligned}$$

Combining this with Lemma A.3 and using the time scaling $t_0 := \frac{2\beta\gamma}{m(\alpha-1)}$, one establishes

$$\|P_t\|_{\alpha \rightarrow \gamma} \leq (1 + 4d + (2\bar{L} + m)R_*^2) \exp\left(\frac{t\gamma(\bar{L}R_*)^2}{4(\alpha-1)/\beta} + \frac{1}{8} \max\left\{\frac{1+4d}{t/t_0-1}, mR_*^2\right\}\right)$$

for all $t \geq t_0 > 0$. Therefore, this completes the proof. $\square$

A.7. Proof of Lemmas 3.9, 3.10 and 3.11 in Section 3.

Proof of Lemma 3.9-(i). For any $0 < \lambda \leq \lambda_1, t \in (n, n+1], n \in \mathbb{N}$, define

$$\Delta_{n,t}^\lambda := \bar{\theta}_n^\lambda - \lambda h_\lambda(\bar{\theta}_n^\lambda)(t - n), \quad \Xi_{n,t}^\lambda := \sqrt{2\lambda\beta^{-1}}(B_t^\lambda - B_n^\lambda). \quad (44)$$

By using (15), one obtains

$$\bar{\theta}_t^\lambda = \Delta_{n,t}^\lambda + \Xi_{n,t}^\lambda. \quad (45)$$

By taking the square of (45) and applying $\mathbb{E}[\cdot | \bar{\theta}_n^\lambda]$ yields

$$\begin{aligned} \mathbb{E}[|\bar{\theta}_t^\lambda|^2 | \bar{\theta}_n^\lambda] &= \mathbb{E}[|\Delta_{n,t}^\lambda|^2 | \bar{\theta}_n^\lambda] + 2\mathbb{E}[\langle \Delta_{n,t}^\lambda, \Xi_{n,t}^\lambda \rangle | \bar{\theta}_n^\lambda] + \mathbb{E}[|\Xi_{n,t}^\lambda|^2 | \bar{\theta}_n^\lambda] \\ &= |\Delta_{n,t}^\lambda|^2 + 2\langle \Delta_{n,t}^\lambda, \mathbb{E}[\Xi_{n,t}^\lambda] \rangle + 2\lambda\beta^{-1}\mathbb{E}[|B_t^\lambda - B_n^\lambda|^2] \\ &= |\Delta_{n,t}^\lambda|^2 + 2\lambda(t-n)d/\beta. \end{aligned} \quad (46)$$

Moreover, by using (9) in Remark 2.10 and Assumption 2, one further calculates, for any $\lambda \in (0, \lambda_1)$,

$$|\Delta_{n,t}^\lambda|^2 = |\bar{\theta}_n^\lambda|^2 - 2\lambda(t-n) \langle \bar{\theta}_n^\lambda, h_\lambda(\bar{\theta}_n^\lambda) \rangle + \lambda^2(t-n)^2 |h_\lambda(\bar{\theta}_n^\lambda)|^2$$

$$\begin{aligned}
&= |\bar{\theta}_n^\lambda|^2 - 2\lambda(t-n) \frac{\langle \bar{\theta}_n^\lambda, h(\bar{\theta}_n^\lambda) \rangle}{1 + \sqrt{\lambda}|\bar{\theta}_n^\lambda|^r} + \lambda^2(t-n)^2 \frac{|h(\bar{\theta}_n^\lambda)|^2}{(1 + \sqrt{\lambda}|\bar{\theta}_n^\lambda|^r)^2} \\
&\leq |\bar{\theta}_n^\lambda|^2 - 2\lambda(t-n) \frac{a^*|\bar{\theta}_n^\lambda|^{r+2} - b^*}{1 + \sqrt{\lambda}|\bar{\theta}_n^\lambda|^r} + \lambda^2(t-n)^2 \frac{|h_\lambda(\bar{\theta}_n^\lambda)|^2}{(1 + \sqrt{\lambda}|\bar{\theta}_n^\lambda|^r)^2} \\
&\leq |\bar{\theta}_n^\lambda|^2 - 2\lambda(t-n) \frac{a^*|\bar{\theta}_n^\lambda|^{r+2} - b^*}{1 + \sqrt{\lambda}|\bar{\theta}_n^\lambda|^r} + \lambda^2(t-n)^2 \frac{2K^2(1 + |\bar{\theta}_n^\lambda|^{2r+2})}{(1 + \sqrt{\lambda}|\bar{\theta}_n^\lambda|^r)^2} \\
&= |\bar{\theta}_n^\lambda|^2 - \lambda(t-n) \frac{a^*|\bar{\theta}_n^\lambda|^{r+2}}{1 + \sqrt{\lambda}|\bar{\theta}_n^\lambda|^r} - \lambda(t-n) \left[\frac{a^*|\bar{\theta}_n^\lambda|^{r+2}}{1 + \sqrt{\lambda}|\bar{\theta}_n^\lambda|^r} - \frac{2\lambda(t-n)K^2|\bar{\theta}_n^\lambda|^{2r+2}}{(1 + \sqrt{\lambda}|\bar{\theta}_n^\lambda|^r)^2} \right] \\
&\quad + 2\lambda(t-n) [b^* + K^2\lambda(t-n)] \\
&= |\bar{\theta}_n^\lambda|^2 - a^*\lambda(t-n)|\bar{\theta}_n^\lambda|^2 T_1^\lambda(\bar{\theta}_n^\lambda) - \lambda(t-n)T_2^\lambda(\bar{\theta}_n^\lambda) + 2\lambda(t-n) [b^* + K^2\lambda(t-n)] \\
&\leq |\bar{\theta}_n^\lambda|^2 - a^*\lambda(t-n)|\bar{\theta}_n^\lambda|^2 T_1^\lambda(\bar{\theta}_n^\lambda) - \lambda(t-n)T_2^\lambda(\bar{\theta}_n^\lambda) + 2\lambda(t-n) (b^* + K^2), \tag{47}
\end{aligned}$$

where the second inequality and last inequality follow from $(a+b)^2 \leq 2(a^2 + b^2)$ and $\lambda(t-n) < 1$ for all $a, b \geq 0$, respectively, and where for all $\theta \in \mathbb{R}^d$, denote by

$$T_1^\lambda(\theta) := \frac{|\theta|^r}{1 + \sqrt{\lambda}|\theta|^r} \tag{48}$$

and moreover, for all $\theta \in \mathbb{R}^d$,

$$T_2^\lambda(\theta) := \frac{a^*|\theta|^{r+2}}{1 + \sqrt{\lambda}|\theta|^r} - \frac{2\lambda(t-n)K^2|\theta|^{2r+2}}{(1 + \sqrt{\lambda}|\theta|^r)^2}.$$

On the one hand, for every $\lambda > 0$, the function $f(s) = s/(1 + \sqrt{\lambda}s)$ is non-decreasing for all $s \geq 0$. Thus, by using $\lambda < \lambda_1 \leq 1$, one obtains

$$T_1^\lambda(\theta) \geq \frac{1}{1 + \sqrt{\lambda}} \geq \frac{1}{2}, \quad \text{for all } |\theta| \geq 1. \tag{49}$$

On the other hand, for all $\theta \in \mathbb{R}^d$ and $\lambda < \lambda_1 \leq (a^*)^2/(4K^4)$, one also obtains

$$\begin{aligned}
T_2^\lambda(\theta) &= \frac{a^*|\theta|^{r+2}}{1 + \sqrt{\lambda}|\theta|^r} - \frac{2\lambda(t-n)K^2|\theta|^{2r+2}}{(1 + \sqrt{\lambda}|\theta|^r)^2} \\
&\geq \frac{a^*\sqrt{\lambda}|\theta|^{2r+2} - 2\lambda(t-n)K^2|\theta|^{2r+2}}{(1 + \sqrt{\lambda}|\theta|^r)^2} \\
&= \frac{(a^* - 2\sqrt{\lambda}K^2)\sqrt{\lambda}|\theta|^{2r+2}}{(1 + \sqrt{\lambda}|\theta|^r)^2} \geq 0
\end{aligned} \tag{50}$$

for all $\theta \in \mathbb{R}^d$. Denote $S_n := \{\omega \in \Omega : |\bar{\theta}_n^\lambda(\omega)| \geq 1\}$. Substituting (49) and (50) into (47) yields, for every $\lambda \in (0, \lambda_1)$,

$$|\Delta_{n,t}^\lambda|^2 \mathbb{1}_{S_n} \leq \left(1 - \frac{a^*}{2}\lambda(t-n)\right) |\bar{\theta}_n^\lambda|^2 \mathbb{1}_{S_n} + 2\lambda(t-n)(b^* + K^2) \mathbb{1}_{S_n}.$$

In addition, by using the definition of $T_1^\lambda(\theta)$ given in (48), one obtains for any $t \in (n, n+1]$,

$$\begin{aligned}
&|\Delta_{n,t}^\lambda|^2 \mathbb{1}_{S_n^c} \\
&\leq \left(1 - \frac{a^*}{2}\lambda(t-n)\right) |\bar{\theta}_n^\lambda|^2 \mathbb{1}_{S_n^c} + 2\lambda(t-n)(b^* + K^2) \mathbb{1}_{S_n^c} + a^*\lambda(t-n) \left(\frac{1}{2} - T_1^\lambda(\bar{\theta}_n^\lambda)\right) |\bar{\theta}_n^\lambda|^2 \mathbb{1}_{S_n^c} \\
&\leq \left(1 - \frac{a^*}{2}\lambda(t-n)\right) |\bar{\theta}_n^\lambda|^2 \mathbb{1}_{S_n^c} + 2\lambda(t-n)(b^* + K^2) \mathbb{1}_{S_n^c} + \frac{a^*}{2}\lambda(t-n) \mathbb{1}_{S_n^c}.
\end{aligned}$$

Combining the above two cases yields

$$|\Delta_{n,t}^\lambda|^2 \leq \left(1 - \frac{a^*}{2}\lambda(t-n)\right) |\bar{\theta}_n^\lambda|^2 + \lambda(t-n) \{2(b^* + K^2) + a^*/2\},$$

Thus, one can conclude from (46) that, for $t \in (n, n+1]$, $n \in \mathbb{N}$ and $\lambda \in (0, \lambda_1)$,

$$\mathbb{E} \left[|\bar{\theta}_t^\lambda|^2 \mid \bar{\theta}_n^\lambda \right] = |\Delta_{n,t}^\lambda|^2 + 2\lambda(t-n)d/\beta \leq \left(1 - \frac{a^*}{2}\lambda(t-n)\right) |\bar{\theta}_n^\lambda|^2 + \lambda(t-n)c_0, \quad (51)$$

where

$$c_0 := 2d\beta^{-1} + 2(b^* + K^2) + a^*/2.$$

Taking expectation on (51) further yields, for all $t \in (n, n+1]$, $n \in \mathbb{N}$ and $\lambda \in (0, \lambda_1)$,

$$\mathbb{E} \left[|\bar{\theta}_t^\lambda|^2 \right] \leq \left(1 - \frac{a^*}{2}\lambda(t-n)\right) \mathbb{E} \left[|\bar{\theta}_n^\lambda|^2 \right] + \lambda(t-n)c_0.$$

This implies, for $t \in (n, n+1]$, $n \in \mathbb{N}$ and $0 < \lambda \leq \lambda_1 \leq 1$, that

$$\begin{aligned} \mathbb{E} \left[|\bar{\theta}_t^\lambda|^2 \right] &\leq \left(1 - \frac{a^*}{2}\lambda(t-n)\right) \left(1 - \frac{a^*\lambda}{2}\right) \mathbb{E} \left[|\bar{\theta}_{n-1}^\lambda|^2 \right] + \lambda_1 c_0 + \lambda c_0 \\ &\leq \left(1 - \frac{a^*}{2}\lambda(t-n)\right) \left(1 - \frac{a^*\lambda}{2}\right)^2 \mathbb{E} \left[|\bar{\theta}_{n-2}^\lambda|^2 \right] + c_0 + \lambda c_0 \left\{1 + \left(1 - \frac{a^*\lambda}{2}\right)\right\} \\ &\leq \left(1 - \frac{a^*}{2}\lambda(t-n)\right) \left(1 - \frac{a^*\lambda}{2}\right)^n \mathbb{E} \left[|\bar{\theta}_0^\lambda|^2 \right] + c_0 \\ &\quad + \lambda c_0 \left\{1 + \left(1 - \frac{a^*\lambda}{2}\right) + \cdots + \left(1 - \frac{a^*\lambda}{2}\right)^{n-1}\right\} \\ &\leq \left(1 - \frac{a^*}{2}\lambda(t-n)\right) \left(1 - \frac{a^*\lambda}{2}\right)^n \mathbb{E} \left[|\bar{\theta}_0^\lambda|^2 \right] + c_0(1 + 2/a^*), \end{aligned}$$

which completes the proof. $\square$

Proof of Lemma 3.9-(ii). For any $p \in [2, \infty) \cap \mathbb{N}$, $0 < \lambda \leq \lambda_2$, $t \in (n, n+1]$ and $n \in \mathbb{N}$, recall that the definition for $\Delta_{n,t}^\lambda$ and $\Xi_{n,t}^\lambda$ in (44). For all $x, y \in \mathbb{R}^d$, one obtains the inequality

$$\sum_{\substack{i+j+k=p \\ \{i \neq p-1\} \cap \{j \neq 1\}}} \frac{p!}{i!j!k!} |x|^{2i} \cdot |2 \langle x, y \rangle|^j \cdot |y|^{2k} \leq \sum_{\{k=0\} \cap \{k \neq 1\}}^{2p} \binom{2p}{k} |x|^{2p-k} |y|^k, \quad (52)$$

for example, see [9, Lemma A.3]. To obtain the $2p$ -th moment estimate of the continuous-time interpolation (15), one also writes

$$\begin{aligned} \mathbb{E} \left[|\bar{\theta}_t^\lambda|^{2p} \mid \bar{\theta}_n^\lambda \right] &= \mathbb{E} \left[\left(|\Delta_{n,t}^\lambda|^2 + 2 \langle \Delta_{n,t}^\lambda, \Xi_{n,t}^\lambda \rangle + |\Xi_{n,t}^\lambda|^2 \right)^p \mid \bar{\theta}_n^\lambda \right] \\ &= \mathbb{E} \left[|\Delta_{n,t}^\lambda|^{2p} \mid \bar{\theta}_n^\lambda \right] + 2p \mathbb{E} \left[|\Delta_{n,t}^\lambda|^{2p-2} \langle \Delta_{n,t}^\lambda, \Xi_{n,t}^\lambda \rangle \mid \bar{\theta}_n^\lambda \right] \\ &\quad + \mathbb{E} \left[\sum_{\substack{k_1+k_2+k_3=p \\ \{k_1 \neq p-1\} \cap \{k_2 \neq 1\} \\ k_1 \neq p}} \frac{p!}{k_1!k_2!k_3!} |\Delta_{n,t}^\lambda|^{2k_1} \left(2 \langle \Delta_{n,t}^\lambda, \Xi_{n,t}^\lambda \rangle \right)^{k_2} |\Xi_{n,t}^\lambda|^{2k_3} \mid \bar{\theta}_n^\lambda \right] \\ &\leq |\Delta_{n,t}^\lambda|^{2p} + 2p \mathbb{E} \left[|\Delta_{n,t}^\lambda|^{2p-2} \langle \Delta_{n,t}^\lambda, \Xi_{n,t}^\lambda \rangle \mid \bar{\theta}_n^\lambda \right] + \sum_{k=2}^{2p} \binom{2p}{k} \mathbb{E} \left[|\Delta_{n,t}^\lambda|^{2p-k} |\Xi_{n,t}^\lambda|^k \mid \bar{\theta}_n^\lambda \right] \\ &= |\Delta_{n,t}^\lambda|^{2p} + \sum_{k=2}^{2p} \binom{2p}{k} \mathbb{E} \left[|\Delta_{n,t}^\lambda|^{2p-k} |\Xi_{n,t}^\lambda|^k \mid \bar{\theta}_n^\lambda \right], \end{aligned}$$

where the inequality follows from (52). This and the fact that $\Xi_{n,t}^\lambda$ is independent of $\Delta_{n,t}^\lambda$ and $\Xi_{n,t}^\lambda$ is independent of $\bar{\theta}_n^\lambda$, $t \in (n, n+1]$, $n \in \mathbb{N}$, yield

$$\mathbb{E} \left[|\bar{\theta}_t^\lambda|^{2p} \mid \bar{\theta}_n^\lambda \right] \leq |\Delta_{n,t}^\lambda|^{2p} + \sum_{k=0}^{2p-2} \binom{2p}{k+2} \mathbb{E} \left[|\Delta_{n,t}^\lambda|^{2p-2-k} |\Xi_{n,t}^\lambda|^{k+2} \mid \bar{\theta}_n^\lambda \right]$$

$$\begin{aligned}
&= |\Delta_{n,t}^\lambda|^{2p} + \sum_{k=0}^{2p-2} \frac{2p(2p-1)}{(k+2)(k+1)} \binom{2p-2}{k} \mathbb{E} \left[\left(|\Delta_{n,t}^\lambda|^{2p-2-k} |\Xi_{n,t}^\lambda|^k \right) |\Xi_{n,t}^\lambda|^2 \mid \bar{\theta}_n^\lambda \right] \\
&\leq |\Delta_{n,t}^\lambda|^{2p} + p(2p-1) \mathbb{E} \left[\left(|\Delta_{n,t}^\lambda| + |\Xi_{n,t}^\lambda| \right)^{2p-2} |\Xi_{n,t}^\lambda|^2 \mid \bar{\theta}_n^\lambda \right] \\
&\leq |\Delta_{n,t}^\lambda|^{2p} + 2^{2p-3} p(2p-1) |\Delta_{n,t}^\lambda|^{2p-2} \mathbb{E} \left[|\Xi_{n,t}^\lambda|^2 \right] + 2^{2p-3} p(2p-1) \mathbb{E} \left[|\Xi_{n,t}^\lambda|^{2p} \right] \\
&\leq |\Delta_{n,t}^\lambda|^{2p} + p(2p-1) 2^{2p-2} \lambda(t-n) \beta^{-1} d |\Delta_{n,t}^\lambda|^{2p-2} \\
&\quad + p(2p-1) 2^{4p-3} (\lambda \beta^{-1} (t-n))^p p! \binom{p-1+d/2}{p}.
\end{aligned} \tag{53}$$

Combining (49), (50) and (47), for any $\lambda \in (0, \lambda_1)$, the proof Lemma 3.9-(i) implies that

$$|\Delta_{n,t}^\lambda|^2 \leq \left(1 - \lambda(t-n) \frac{a^* |\bar{\theta}_n^\lambda|^r}{1 + \sqrt{\lambda} |\bar{\theta}_n^\lambda|^r} \right) |\bar{\theta}_n^\lambda|^2 + \lambda(t-n) (2b^* + 2K^2). \tag{54}$$

Moreover, for any $0 < \lambda \leq 1/(a^*)^2$, one also obtains

$$1 > 1 - \lambda(t-n) \frac{a^* |\bar{\theta}_n^\lambda|^r}{1 + \sqrt{\lambda} |\bar{\theta}_n^\lambda|^r} \geq 1 - \lambda \left(\frac{a^* |\bar{\theta}_n^\lambda|^r}{\sqrt{\lambda} |\bar{\theta}_n^\lambda|^r} \right) = 1 - \sqrt{\lambda} a^* \geq 0.$$

By taking the p -th power on both sides in (54), then it yields, for any $\lambda \in (0, \lambda_2)$,

$$\begin{aligned}
|\Delta_{n,t}^\lambda|^{2p} &\leq \sum_{k=0}^p \binom{n}{k} \left\{ \left(1 - \lambda(t-n) \frac{a^* |\bar{\theta}_n^\lambda|^r}{1 + \sqrt{\lambda} |\bar{\theta}_n^\lambda|^r} \right) |\bar{\theta}_n^\lambda|^2 \right\}^{p-k} \lambda^k (t-n)^k (2b^* + 2K^2)^k \\
&\leq \left(1 - \lambda(t-n) \frac{a^* |\bar{\theta}_n^\lambda|^r}{1 + \sqrt{\lambda} |\bar{\theta}_n^\lambda|^r} \right) |\bar{\theta}_n^\lambda|^{2p} + \sum_{k=1}^p \binom{n}{k} \lambda^k (t-n)^k (2b^* + 2K^2)^k |\bar{\theta}_n^\lambda|^{2(p-k)} \\
&= \left(1 - \lambda(t-n) \frac{a^* |\bar{\theta}_n^\lambda|^r}{2(1 + \sqrt{\lambda} |\bar{\theta}_n^\lambda|^r)} \right) |\bar{\theta}_n^\lambda|^{2p} - J_1^\lambda(\bar{\theta}_n^\lambda),
\end{aligned} \tag{55}$$

where for every $\theta \in \mathbb{R}^d$, define

$$J_1^\lambda(\theta) := \lambda(t-n) \frac{a^* |\theta|^{r+2p}}{2(1 + \sqrt{\lambda} |\theta|^r)} - \sum_{k=1}^p \binom{n}{k} (2b^* + 2K^2)^k \lambda^k (t-n)^k |\theta|^{2(p-k)}.$$

On the other hand, for any $\theta \in \mathbb{R}^d$ and $\lambda \in (0, \lambda_2)$, the term $J_1^\lambda(\theta)$ can be bounded below by

$$\begin{aligned}
&J_1^\lambda(\theta) \\
&\geq \lambda(t-n) \frac{a^* |\theta|^{r+2p}}{2(1 + \sqrt{\lambda} |\theta|^r)} - \sum_{k=1}^p \binom{n}{k} \lambda^k (t-n)^k (1 + 2b^* + 2K^2)^k |\theta|^{2(p-k)} \\
&\geq \lambda(t-n) \frac{a^* |\theta|^{r+2p}}{2(1 + \sqrt{\lambda} |\theta|^r)} - \binom{p}{\lfloor p/2 \rfloor} \lambda(t-n) (1 + 2b^* + 2K^2)^p \sum_{k=1}^p |\theta|^{2(p-k)} \cdot \frac{1 + \sqrt{\lambda} |\theta|^r}{1 + \sqrt{\lambda} |\theta|^r} \\
&\geq \lambda(t-n) \frac{a^* |\theta|^{r+2p}}{2(1 + \sqrt{\lambda} |\theta|^r)} - \binom{p}{\lfloor p/2 \rfloor} \lambda(t-n) (1 + 2b^* + 2K^2)^p \sum_{k=1}^p |\theta|^{2(p-k)} \cdot \frac{1 + |\theta|^r}{1 + \sqrt{\lambda} |\theta|^r} \\
&= \frac{\lambda(t-n)}{1 + \sqrt{\lambda} |\theta|^r} \sum_{k=1}^p \left[\frac{a^*}{4p} |\theta|^{r+2p} - \binom{p}{\lfloor p/2 \rfloor} (1 + 2b^* + 2K^2)^p |\theta|^{2(p-k)} \right] \\
&\quad + \frac{\lambda(t-n)}{1 + \sqrt{\lambda} |\theta|^r} \sum_{k=1}^p \left[\frac{a^*}{4p} |\theta|^{r+2p} - \binom{p}{\lfloor p/2 \rfloor} (1 + 2b^* + 2K^2)^p |\theta|^{r+2(p-k)} \right].
\end{aligned} \tag{56}$$

To obtain an explicit form of such a constant, one notes that for every $1 \leq k \leq p$,

$$\frac{a^*}{4p} |\theta|^{r+2p} - \binom{p}{\lfloor p/2 \rfloor} (1 + 2b^* + 2K^2)^p |\theta|^{2(p-k)} > 0$$

$$\iff |\theta| > \left[\frac{4p \binom{p}{\lfloor p/2 \rfloor} (1 + 2b^* + 2K^2)^p}{a^*} \right]^{1/(2k+r)}$$

and

$$\begin{aligned} & \frac{a^*}{4p} |\theta|^{r+2p} - \binom{p}{\lfloor p/2 \rfloor} (1 + 2b^* + 2K^2)^p |\theta|^{r+2(p-k)} > 0 \\ \iff & |\theta| > \left[\frac{4p \binom{p}{\lfloor p/2 \rfloor} (1 + 2b^* + 2K^2)^p}{a^*} \right]^{1/(2k)}. \end{aligned}$$

Denote by

$$M_1(p) := \frac{4p \binom{p}{\lfloor p/2 \rfloor} (1 + 2b^* + 2K^2)^p}{\min \{1, a^*\}}, \quad \forall p \in [2, \infty) \cap \mathbb{N}.$$

In particular, one also obtains

$$M_1(p) \geq \max_{1 \leq k \leq p} \max \left\{ \left[\frac{4p \binom{p}{\lfloor \frac{p}{2} \rfloor} (1 + 2b^* + 2K^2)^p}{a^*} \right]^{\frac{1}{2k+r}}, \left[\frac{4p \binom{p}{\lfloor \frac{p}{2} \rfloor} (1 + 2b^* + 2K^2)^p}{a^*} \right]^{\frac{1}{2k}} \right\}. \quad (57)$$

Combining (56) and (57), for every $\lambda \in (0, \lambda_2)$ and $\theta \in \mathbb{R}^d$ such that $|\theta| > M_1(p)$, it yields that

$$J_1^\lambda(\theta) > 0, \quad \text{for all } |\theta| > M_1(p). \quad (58)$$

Therefore, denote by $S_{n, M_1(p)} := \{\omega \in \Omega : |\bar{\theta}_n^\lambda(\omega)| > M_1(p)\}$. Then by using (58) and (55), it follows that

$$\begin{aligned} |\Delta_{n,t}^\lambda|^{2p} \mathbb{1}_{S_{n, M_1(p)}} & \leq \left(1 - \lambda(t-n) \frac{a^* |\bar{\theta}_n^\lambda|^r}{2(1 + \sqrt{\lambda} |\bar{\theta}_n^\lambda|^r)} \right) |\bar{\theta}_n^\lambda|^{2p} \mathbb{1}_{S_{n, M_1(p)}} \\ & \leq (1 - \lambda(t-n) a^* \kappa(p)) |\bar{\theta}_n^\lambda|^{2p} \mathbb{1}_{S_{n, M_1(p)}}, \end{aligned} \quad (59)$$

where we used the fact that $f(s) = s/(1 + \sqrt{\lambda}s)$ is non-decreasing on $[0, +\infty)$ and where

$$\kappa(p) := \frac{|M_1(p)|^r}{2(1 + |M_1(p)|^r)}, \quad \text{for all } p \in \mathbb{N}.$$

Furthermore, for any $0 < \lambda < \lambda_2$, one obtains

$$\begin{aligned} & |\Delta_{n,t}^\lambda|^{2p} \mathbb{1}_{S_{n, M_1(p)}^c} \\ & \leq |\bar{\theta}_n^\lambda|^{2p} \mathbb{1}_{S_{n, M_1(p)}^c} - J_1^\lambda(\bar{\theta}_n^\lambda) \mathbb{1}_{S_{n, M_1(p)}^c} \\ & = (1 - \lambda(t-n) a^* \kappa(p)) |\bar{\theta}_n^\lambda|^{2p} \mathbb{1}_{S_{n, M_1(p)}^c} + \lambda(t-n) a^* \kappa(p) |\bar{\theta}_n^\lambda|^{2p} \mathbb{1}_{S_{n, M_1(p)}^c} - J_1^\lambda(\bar{\theta}_n^\lambda) \mathbb{1}_{S_{n, M_1(p)}^c} \\ & \leq (1 - \lambda(t-n) a^* \kappa(p)) |\bar{\theta}_n^\lambda|^{2p} \mathbb{1}_{S_{n, M_1(p)}^c} + \lambda(t-n) a^* \kappa(p) |M_1(p)|^{2p} \mathbb{1}_{S_{n, M_1(p)}^c} \\ & \quad + \lambda(t-n) \sum_{k=1}^p \binom{n}{k} (2b^* + 2K^2)^k |M_1(p)|^{2(p-k)} \mathbb{1}_{S_{n, M_1(p)}^c} \\ & = (1 - \lambda(t-n) a^* \kappa(p)) |\bar{\theta}_n^\lambda|^{2p} \mathbb{1}_{S_{n, M_1(p)}^c} + \lambda(t-n) c_1(p) \mathbb{1}_{S_{n, M_1(p)}^c}, \end{aligned} \quad (60)$$

where

$$c_1(p) := a^* \kappa(p) |M_1(p)|^{2p} + \sum_{k=1}^p \binom{n}{k} (2b^* + 2K^2)^k |M_1(p)|^{2(p-k)}.$$

Combining the inequalities (59) and (60) yields, for every $p \in \mathbb{N}$ and $\lambda \in (0, \lambda_2)$,

$$|\Delta_{n,t}^\lambda|^{2p} \leq (1 - \lambda(t-n) a^* \kappa(p)) |\bar{\theta}_n^\lambda|^{2p} + \lambda(t-n) c_1(p). \quad (61)$$

By using (61) and the fact that $a^*/2 + 2b^* + 2K^2 < c_1(1)$, it yields that for every $p \in [2, \infty) \cap \mathbb{N}$, $\lambda \in (0, \lambda_2)$,

$$|\Delta_{n,t}^\lambda|^{2p-2} \leq |\bar{\theta}_n^\lambda|^{2p-2} + \lambda(t-n) c_1(p-1). \quad (62)$$

Inserting (61) and (62) into (53) yields, for every $p \in [2, \infty) \cap \mathbb{N}$, $\lambda \in (0, \lambda_2)$, $t \in (n, n+1]$ and $n \in \mathbb{N}$,

$$\begin{aligned} & \mathbb{E} \left[|\bar{\theta}_t^\lambda|^{2p} \mid \bar{\theta}_n^\lambda \right] \\ & \leq (1 - \lambda(t-n)a^*\kappa(p))|\bar{\theta}_n^\lambda|^{2p} + p(2p-1)\lambda\beta^{-1}d(t-1)|\bar{\theta}_n^\lambda|^{2p-2} \\ & \quad + p(2p-1)2^{2p-2}\lambda^2\beta^{-1}d(t-n)^2c_1(p-1) + \lambda(t-n)c_1(p) \\ & \quad + p(2p-1)2^{4p-3}(\lambda\beta^{-1}(t-n))^p p! \binom{p-1+d/2}{p} \\ & \leq (1 - \lambda(t-n)a^*\kappa(p))|\bar{\theta}_n^\lambda|^{2p} + p(2p-1)\lambda\beta^{-1}d(t-1)|\bar{\theta}_n^\lambda|^{2p-2} + \lambda(t-n)c_2(p), \end{aligned} \quad (63)$$

where

$$c_2(p) := c_1(p) + p(2p-1)2^{2p-2}\beta^{-1}c_1(p-1)d + p(2p-1)2^{4p-3}\beta^{-p}p! \binom{p-1+d/2}{p}.$$

Observe that, for all $\theta \in \mathbb{R}^d$,

$$\frac{a^*\kappa(p)}{2}|\theta|^{2p} - p(2p-1)2^{2p-2}\beta^{-1}d|\theta|^{2p-2} > 0 \iff |\theta| > M_2(p),$$

where

$$M_2(p) := \left(\frac{p(2p-1)2^{2p-1}\beta^{-1}d}{a^*\kappa(p)} \right)^{1/2}.$$

Denote by $S_{n,M_2(p)} := \{\omega \in \Omega : |\bar{\theta}_n^\lambda(\omega)| > M_2(p)\}$, it follows by (63),

$$\mathbb{E} \left[|\bar{\theta}_t^\lambda|^{2p} \mathbb{1}_{S_{n,M_2(p)}} \mid \bar{\theta}_n^\lambda \right] \leq \left(1 - \frac{\lambda(t-n)a^*\kappa(p)}{2} \right) |\bar{\theta}_n^\lambda|^{2p} \mathbb{1}_{S_{n,M_2(p)}} + \lambda(t-n)c_2(p) \mathbb{1}_{S_{n,M_2(p)}}. \quad (64)$$

Furthermore, one also obtains

$$\mathbb{E} \left[|\bar{\theta}_t^\lambda|^{2p} \mathbb{1}_{S_{n,M_2(p)}^c} \mid \bar{\theta}_n^\lambda \right] \leq \left(1 - \frac{\lambda(t-n)a^*\kappa(p)}{2} \right) |\bar{\theta}_n^\lambda|^{2p} \mathbb{1}_{S_{n,M_2(p)}^c} + \lambda(t-n)c_3(p) \mathbb{1}_{S_{n,M_2(p)}}, \quad (65)$$

where

$$c_3(p) := c_2(p) + p(2p-1)2^{2p-2}d\beta^{-1}|M_2(p)|^{2p-2}.$$

Therefore, combining (64) and (65) yields, for all $p \in [2, \infty) \cap \mathbb{N}$, $\lambda \in (0, \lambda_2)$,

$$\mathbb{E} \left[|\bar{\theta}_t^\lambda|^{2p} \mid \bar{\theta}_n^\lambda \right] \leq \left(1 - \frac{\lambda(t-n)a^*\kappa(p)}{2} \right) |\bar{\theta}_n^\lambda|^{2p} + \lambda(t-n)c_3(p), \quad (66)$$

where the constant $c_3(p)$ is given by

$$\kappa(p) := \frac{|M_1(p)|^r}{2(1+|M_1(p)|^r)}, \quad M_1(p) := \frac{4p \binom{p}{\lfloor p/2 \rfloor} (1+2b^*+2K^2)^p}{\min\{1, a^*\}}, \quad (67)$$

$$c_3(p) := c_2(p) + p(2p-1)2^{2p-2}d\beta^{-1}|M_2(p)|^{2p-2}, \quad M_2(p) := \left(\frac{p(2p-1)2^{2p-1}d\beta^{-1}}{a^*\kappa(p)} \right)^{1/2}, \quad (68)$$

$$c_2(p) := c_1(p) + p(2p-1)2^{2p-2}d\beta^{-1}c_1(p-1) + p(2p-1)2^{4p-3}\beta^{-p}p! \binom{p-1+d/2}{p},$$

$$c_1(p) := a^*\kappa(p)|M_1(p)|^{2p} + \sum_{k=1}^p \binom{n}{k} (2b^*+2K^2)^k |M_1(p)|^{2(p-k)}.$$

Therefore, by noting $\kappa(p) \geq \kappa(2)$ for all $p \in [2, \infty) \cap \mathbb{N}$ and by using similar argument as in the proof of Lemma 3.9-(i), one can conclude that from (66) and for $p \in [2, \infty) \cap \mathbb{N}$, $\lambda \in (0, \lambda_2)$, $t \in (n, n+1]$, $n \in \mathbb{N}$,

$$\mathbb{E} \left[|\bar{\theta}_t^\lambda|^{2p} \right] \leq \left(1 - \lambda(t-n) \frac{a^*\kappa(2)}{2} \right) \left(1 - \frac{\lambda a^*\kappa(2)}{2} \right)^n \mathbb{E} \left[|\bar{\theta}_0^\lambda|^{2p} \right] + c_3(p) \left(1 + \frac{2}{a^*\kappa(2)} \right).$$

This completes the proof. $\square$

Proof of Lemma 3.10. For any $t \in \mathbb{R}_+$, by using the continuous-time interpolation of the TUSLA given in (15), one obtains

$$\begin{aligned} \mathbb{E} \left[|\bar{\theta}_t^\lambda - \bar{\theta}_{[t]}^\lambda|^4 \right] &= \mathbb{E} \left[\left| -\lambda \int_{[t]}^t h_\lambda(\bar{\theta}_{[s]}^\lambda) ds + \sqrt{2\lambda\beta^{-1}} \int_{[t]}^t dB_s^\lambda \right|^4 \right] \\ &\leq 8\lambda^4 \mathbb{E} \left[|h(\bar{\theta}_{[t]}^\lambda)|^4 \right] + 32\lambda^2 \beta^{-2} \mathbb{E} \left[\sum_{i=1}^d |B_1^{i,\lambda}|^4 + 2 \sum_{1 \leq i < j \leq d} |B_1^{i,\lambda}|^2 \cdot |B_1^{j,\lambda}|^2 \right] \\ &= 8\lambda^4 \mathbb{E} \left[|h(\bar{\theta}_{[t]}^\lambda)|^4 \right] + 32\lambda^2 \beta^{-2} d(d+2), \end{aligned}$$

where the first inequality follows from $[s] = [t]$ for any $s \in [[t], t]$ and $(B_1^{1,\lambda}, \dots, B_1^{d,\lambda})$ be a standard d -dimensional Gaussian random variables. Moreover, by applying Assumption 2 and Lemma 3.9-(ii) yield, for any $\lambda \in (0, \lambda_2)$,

$$\begin{aligned} \mathbb{E} \left[|\bar{\theta}_t^\lambda - \bar{\theta}_{[t]}^\lambda|^4 \right] &\leq 64\lambda^4 K^4 \mathbb{E} \left[|\bar{\theta}_{[t]}^\lambda|^{4r+4} \right] + 64\lambda^4 K^4 + 32d(d+2)\lambda^2 \beta^{-2} \\ &\leq 64\lambda^4 K^4 \left\{ e^{-\lambda a^* \kappa(2) [t]/2} \mathbb{E} \left[|\theta_0|^{4r+4} \right] + c_3(2r+2) \left(1 + \frac{2}{a^* \kappa(2)} \right) \right\} \\ &\quad + 64\lambda^4 K^4 + 32d(d+2)\lambda^2 \beta^{-2} \\ &\leq \lambda^2 \left\{ 64K^4 \exp \left(-\frac{\lambda a^* \kappa(2)}{2} [t] \right) \mathbb{E} \left[|\theta_0|^{4r+4} \right] + \bar{C}_1 \right\}, \end{aligned}$$

where the second inequality holds due to $1 - x \leq e^{-x}$ for any $x \in \mathbb{R}$, and where

$$\bar{C}_1 := 64K^4 \left\{ 1 + c_3(2r+2) \left(1 + \frac{2}{a^* \kappa(2)} \right) \right\} + 32d(d+2)\beta^{-2}, \quad (69)$$

with K given in Assumption 2, and $c_3(2r+2), \kappa(2)$ given in Lemma 3.9. $\square$

Proof of Lemma 3.11. On the one hand, by the definition of $h_\lambda(\theta)$ and Assumption 2, then it yields, for any $n \in \mathbb{N}$,

$$\begin{aligned} |h(\bar{\theta}_n^\lambda) - h_\lambda(\bar{\theta}_n^\lambda)|^2 &= \left| h(\bar{\theta}_n^\lambda) - \frac{h(\bar{\theta}_n^\lambda)}{1 + \sqrt{\lambda} |\bar{\theta}_n^\lambda|^r} \right|^2 = |h(\bar{\theta}_n^\lambda)|^2 \cdot \frac{\lambda |\bar{\theta}_n^\lambda|^{2r}}{(1 + \sqrt{\lambda} |\bar{\theta}_n^\lambda|^r)^2} \\ &\leq K^2 \left(1 + |\bar{\theta}_n^\lambda|^{1+r} \right)^2 \cdot \lambda \left(1 + |\bar{\theta}_n^\lambda|^{2+2r} \right) \\ &\leq 4\lambda K^2 \left(1 + |\bar{\theta}_n^\lambda|^{4+4r} \right), \end{aligned}$$

where the last inequality follows from $(a+b)^2 \leq 2(a^2 + b^2)$ twice for any $a, b \in \mathbb{R}_+$. Furthermore, by using Lemma 3.9-(ii), the following estimate can be obtained:

$$\mathbb{E} \left[|h(\bar{\theta}_n^\lambda) - h_\lambda(\bar{\theta}_n^\lambda)|^2 \right] \leq 4\lambda K^2 \left(1 + \mathbb{E} \left[|\bar{\theta}_n^\lambda|^{4r+4} \right] \right) \leq \lambda C_1^\dagger,$$

where

$$C_1^\dagger := 4K^2 \left\{ 1 + \mathbb{E} \left[|\theta_0|^{4(r+1)} \right] + c_3(2r+2) \left(1 + \frac{2}{a^* \kappa(2)} \right) \right\}. \quad (70)$$

On the other hand, by using Assumption 2 and Cauchy-Schwarz inequality, one obtains, for any $t \in [n, n+1)$,

$$\begin{aligned} \mathbb{E} \left[|h(\bar{\theta}_t^\lambda) - h(\bar{\theta}_n^\lambda)|^2 \right] &\leq L^2 \mathbb{E} \left[\left(1 + |\bar{\theta}_t^\lambda| + |\bar{\theta}_n^\lambda| \right)^{2r} |\bar{\theta}_t^\lambda - \bar{\theta}_n^\lambda|^2 \right] \\ &\leq L^2 \left(\mathbb{E} \left[\left(1 + |\bar{\theta}_t^\lambda| + |\bar{\theta}_n^\lambda| \right)^{4r} \right] \right)^{1/2} \left(\mathbb{E} \left[|\bar{\theta}_t^\lambda - \bar{\theta}_n^\lambda|^4 \right] \right)^{1/2} \\ &\leq 3^{2r-1/2} L^2 \left(\mathbb{E} \left[1 + |\bar{\theta}_t^\lambda|^{4r} + |\bar{\theta}_n^\lambda|^{4r} \right] \right)^{1/2} \left(\mathbb{E} \left[|\bar{\theta}_t^\lambda - \bar{\theta}_n^\lambda|^4 \right] \right)^{1/2}. \quad (71) \end{aligned}$$

Combining Lemma 3.9-(ii) and Lemma 3.10, it yields that

$$\mathbb{E} \left[1 + |\bar{\theta}_t^\lambda|^{4r} + |\bar{\theta}_n^\lambda|^{4r} \right] \leq C^*, \quad (72)$$

where

$$C^* := 1 + 2c_3(2r) + \frac{4c_3(2r)}{a^* \kappa(2)} + 2\mathbb{E}[|\theta_0|^{4r}] \quad (73)$$

and

$$\begin{aligned} \left(\mathbb{E}[|\bar{\theta}_t^\lambda - \bar{\theta}_n^\lambda|^4] \right)^{1/2} &\leq \lambda \left\{ 8K^2 \exp\left(-\frac{n\lambda a^* \kappa(2)}{4}\right) (\mathbb{E}[|\theta_0|^{4r+4}])^{1/2} + \sqrt{\bar{C}_1} \right\} \\ &\leq \lambda \left(8K^2 \sqrt{\mathbb{E}[|\theta_0|^{4r+4}]} + \sqrt{\bar{C}_1} \right). \end{aligned} \quad (74)$$

Hence, by substituting (72) and (74) into (71), it yields that

$$\mathbb{E}[|h(\bar{\theta}_t^\lambda) - h(\bar{\theta}_n^\lambda)|^2] \leq \lambda C_2^\dagger,$$

where

$$C_2^\dagger := 3^{2r-1/2} L^2 \sqrt{C^*} \left(8K^2 \sqrt{\mathbb{E}[|\theta_0|^{4r+4}]} + \sqrt{\bar{C}_1} \right), \quad (75)$$

with constants $C^*, \bar{C}_1$ are explicitly given in (73) and (69), respectively. This completes the proof. $\square$

A.8. Proof of Lemma 3.12 and Theorem 3.15 in Section 3.

Proof of Lemma 3.12. On the one hand, one gets, by applying the definition of expectation,

$$\mathbb{E} \left[\nabla_x \cdot \left\{ \pi_{\bar{\theta}_t^\lambda | \bar{\theta}_n^\lambda}(x | \bar{\theta}_n^\lambda) h_\lambda(\bar{\theta}_n^\lambda) \right\} \right] = \int_{\mathbb{R}^d} \nabla_x \cdot \left\{ \pi_{\bar{\theta}_t^\lambda | \bar{\theta}_n^\lambda}(x | y) h_\lambda(y) \right\} \pi_n(y) dy. \quad (76)$$

The interchange of differentiation and integration is justified by Lemmas A.5 and A.7 in [35], which provide the necessary regularity conditions for differentiation under the integral sign with respect to the spatial variable $x \in \mathbb{R}^d$, then

$$\int_{\mathbb{R}^d} \nabla_x \cdot \left\{ \pi_{\bar{\theta}_t^\lambda | \bar{\theta}_n^\lambda}(x | y) h_\lambda(y) \right\} \pi_n(y) dy = \nabla_x \cdot \left\{ \int_{\mathbb{R}^d} \pi_n(y) \pi_{\bar{\theta}_t^\lambda | \bar{\theta}_n^\lambda}(x | y) h_\lambda(y) dy \right\}. \quad (77)$$

On the other hand, an application of Bayes's theorem yields

$$\begin{aligned} \nabla_x \cdot \left\{ \int_{\mathbb{R}^d} \pi_n(y) \pi_{\bar{\theta}_t^\lambda | \bar{\theta}_n^\lambda}(x | y) h_\lambda(y) dy \right\} &= \nabla_x \cdot \left\{ \int_{\mathbb{R}^d} \pi_t(x) \pi_{\bar{\theta}_n^\lambda | \bar{\theta}_t^\lambda}(y | x) h_\lambda(y) dy \right\} \\ &= \nabla_x \cdot \left\{ \pi_t(x) \mathbb{E} \left[h_\lambda(\bar{\theta}_n^\lambda) | \bar{\theta}_t^\lambda = x \right] \right\}. \end{aligned} \quad (78)$$

Combining these results through substitution of (77) and (78) into (76), one obtains

$$\mathbb{E} \left[\nabla_x \cdot \left\{ \pi_{\bar{\theta}_t^\lambda | \bar{\theta}_n^\lambda}(x | \bar{\theta}_n^\lambda) h_\lambda(\bar{\theta}_n^\lambda) \right\} \right] = \nabla_x \cdot \left\{ \pi_t(x) \mathbb{E} \left[h_\lambda(\bar{\theta}_n^\lambda) | \bar{\theta}_t^\lambda = x \right] \right\}.$$

The remaining claims follow directly from Lemmas 7.1 and 7.2 in [35], which completes the proof. $\square$

Proof of Theorem 3.15. By using Proposition 3.14, one obtains, for any $t \in [n, n+1]$,

$$\begin{aligned} &\frac{d}{dt} \text{KL}(\pi_t | \pi_\beta) \\ &= -\lambda \left\{ \int_{\mathbb{R}^d} \left\langle \pi_t(x) \mathbb{E} \left[h_\lambda(\bar{\theta}_n^\lambda) | \bar{\theta}_t^\lambda = x \right] + \frac{\nabla \pi_t(x)}{\beta}, \nabla \log \pi_t(x) - \nabla \log \pi_\beta(x) \right\rangle dx \right\} \\ &= -\lambda \left\{ \int_{\mathbb{R}^d} \pi_t(x) \left\langle \mathbb{E} \left[h_\lambda(\bar{\theta}_n^\lambda) | \bar{\theta}_t^\lambda = x \right] + \frac{\nabla \log \pi_t(x)}{\beta}, \nabla \log \pi_t(x) - \nabla \log \pi_\beta(x) \right\rangle dx \right\} \\ &= -\lambda \left\{ \int_{\mathbb{R}^d} \pi_t(x) \left\langle \mathbb{E} \left[h_\lambda(\bar{\theta}_n^\lambda) | \bar{\theta}_t^\lambda = x \right] + \frac{\nabla \log \pi_\beta(x)}{\beta}, \nabla \log \pi_t(x) - \nabla \log \pi_\beta(x) \right\rangle dx \right\} \\ &\quad - \beta^{-1} \lambda \int_{\mathbb{R}^d} |\nabla \log \pi_t(x) - \nabla \log \pi_\beta(x)|^2 \pi_t(x) dx \\ &= -\lambda \left\{ \beta^{-1} I_{\pi_\beta}(\pi_t) + \int_{\mathbb{R}^d} \pi_t(x) \left\langle \mathbb{E} \left[h_\lambda(\bar{\theta}_n^\lambda) - h(x) | \bar{\theta}_t^\lambda = x \right], \nabla \log \pi_t(x) - \nabla \log \pi_\beta(x) \right\rangle dx \right\} \\ &= -\lambda \left\{ \beta^{-1} I_{\pi_\beta}(\pi_t) + \int_{\mathbb{R}^d} \pi_t(x) \left\langle \mathbb{E} \left[h_\lambda(\bar{\theta}_n^\lambda) - h(\bar{\theta}_t^\lambda) | \bar{\theta}_t^\lambda = x \right], \nabla \log \pi_t(x) - \nabla \log \pi_\beta(x) \right\rangle dx \right\} \\ &\leq -(\beta^{-1} \lambda) I_{\pi_\beta}(\pi_t) + \lambda \left\{ \beta \int_{\mathbb{R}^d} \left| \mathbb{E} \left[h_\lambda(\bar{\theta}_n^\lambda) - h(\bar{\theta}_t^\lambda) | \bar{\theta}_t^\lambda = x \right] \right|^2 \pi_t(x) dx + \left(\frac{1}{4\beta} \right) I_{\pi_\beta}(\pi_t) \right\} \end{aligned}$$

$$\begin{aligned}
&= - \left(\frac{3\lambda}{4\beta} \right) I_{\pi_\beta}(\pi_t) + \lambda\beta \int_{\mathbb{R}^d} \left| \int_{\mathbb{R}^d} \pi_{\bar{\theta}_n^\lambda | \bar{\theta}_t^\lambda}(y | x) (h_\lambda(y) - h(x)) dy \right|^2 \pi_t(x) dx \\
&\leq - \left(\frac{3\lambda}{4\beta} \right) I_{\pi_\beta}(\pi_t) + \lambda\beta \int_{\mathbb{R}^d} \pi_t(x) \left\{ \int_{\mathbb{R}^d} \pi_{\bar{\theta}_n^\lambda | \bar{\theta}_t^\lambda}(y | x) |h_\lambda(y) - h(x)|^2 dy \right\} dx \\
&= - \left(\frac{3\lambda}{4\beta} \right) I_{\pi_\beta}(\pi_t) + (\lambda\beta) \mathbb{E} \left[|h_\lambda(\bar{\theta}_n^\lambda) - h(\bar{\theta}_t^\lambda)|^2 \right],
\end{aligned}$$

where the fourth equality follows from the definition of relative Fisher information π_t with respect to π_β , the first inequality was obtained using $\langle a, b \rangle \leq \frac{|a|^2}{4\beta} + \beta|b|^2$ for all $a, b \in \mathbb{R}^d, \beta > 0$, and the last inequality holds due to the Jensen inequality. This completes the proof. $\square$

A.9. Complementary details to Section 4.

Proof of Proposition 4.1. To check Assumptions 2 and 3. For the Gaussian Mixture model, one obtains

$$h(\theta) = \theta - \dot{m} + \frac{2\dot{m}}{1 + e^{2\dot{m}^\top \theta}}$$

for all $\theta \in \mathbb{R}^d$ and $|\dot{m}| > 1$. We introduce the following notation:

$$x := 2 \langle \dot{m}, \theta_1 \rangle, \quad y := 2 \langle \dot{m}, \theta_2 \rangle$$

for all $\theta_1, \theta_2 \in \mathbb{R}^d$. Combining the inequality $1 + u \leq e^u$ for all $u \in \mathbb{R}$ and the Cauchy-Schwarz inequality, we have

$$\begin{aligned}
|h(\theta_1) - h(\theta_2)| &= |\theta_1 - \theta_2 - 2\dot{m} \{ (1 + e^x)^{-1} - (1 + e^y)^{-1} \}| \\
&\leq |\theta_1 - \theta_2| + 2|\dot{m}| \frac{|e^x - e^y|}{(1 + e^x)(1 + e^y)} \\
&= |\theta_1 - \theta_2| + 2|\dot{m}| \frac{e^{\max\{x, y\}} \cdot (1 - \exp(-|x - y|))}{(1 + e^x)(1 + e^y)} \\
&\leq |\theta_1 - \theta_2| + 2|\dot{m}| \{1 - \exp(-|x - y|)\} \\
&\leq |\theta_1 - \theta_2| + 2|\dot{m}||x - y| \leq (1 + 4|\dot{m}|^2)|\theta_1 - \theta_2|.
\end{aligned}$$

Furthermore, one also gets

$$|h(\theta)| \leq |\theta| + \left| 1 - \frac{e^{2\dot{m}^\top \theta} - 1}{e^{2\dot{m}^\top \theta} + 1} \right| \cdot |\dot{m}| \leq |\theta| + |\dot{m}| \leq \max\{1, |\dot{m}|\}(1 + |\theta|)$$

for all $\theta \in \mathbb{R}^d$. Therefore, the above calculation implies that the Assumption 2 holds with

$$L := 1 + 4|\dot{m}|^2, \quad K := \max\{1, |\dot{m}|\}, \quad r = 0.$$

On the other hand, by the Cauchy-Schwarz inequality, for any $\theta \in \mathbb{R}^d$

$$\begin{aligned}
\langle \theta, h(\theta) \rangle &= |\theta|^2 - \frac{e^{2\dot{m}^\top \theta} - 1}{e^{2\dot{m}^\top \theta} + 1} \cdot \langle \dot{m}, \theta \rangle \geq |\theta|^2 - |\langle \dot{m}, \theta \rangle|, \\
&\geq \frac{|\theta|^2}{2} - \left(|\dot{m}\theta| - \frac{|\theta|^2}{2} \right) \mathbb{1}_{\{|\theta| > 2|\dot{m}|\}} - \left(|\dot{m}\theta| - \frac{|\theta|^2}{2} \right) \mathbb{1}_{\{|\theta| \leq 2|\dot{m}|\}}, \\
&\geq \frac{|\theta|^2}{2} - \left(2|\dot{m}|^2 - \frac{|\theta|^2}{2} \right) \mathbb{1}_{\{|\theta| \leq 2|\dot{m}|\}}, \\
&\geq \frac{|\theta|^2}{2} - 2|\dot{m}|^2,
\end{aligned}$$

where the third inequality follows from $|\dot{m}| \cdot |\theta| - |\theta|^2/2 < 0$ when θ satisfies $|\theta| > 2|\dot{m}|$. Thus, the Assumption 3 holds with

$$\bar{a} := 1/2, \quad \bar{b} = 2, \quad r := 0,$$

which completes the proof. $\square$

Proof of Proposition 4.2. The gradient of the double-well potential is given by

$$h(\theta) = \theta(|\theta|^2 - 1)$$

for all $\theta \in \mathbb{R}^d$. For any $\theta_1, \theta_2 \in \mathbb{R}^d$, it yields that

$$\begin{aligned} |h(\theta_1) - h(\theta_2)| &= |(\theta_1|\theta_1|^2 - \theta_2|\theta_2|^2) - (\theta_1 - \theta_2)| \\ &= |(\theta_1|\theta_1|^2 - \theta_2|\theta_1|^2 + \theta_2|\theta_1|^2 - \theta_2|\theta_2|^2) - (\theta_1 - \theta_2)| \\ &\leq |\theta_1|^2 \cdot |\theta_1 - \theta_2| + |\theta_2|(|\theta_1| + |\theta_2|)|\theta_1 - \theta_2| + |\theta_1 - \theta_2| \\ &\leq (|\theta_1| + |\theta_2|)^2 |\theta_1 - \theta_2| + |\theta_1 - \theta_2| \\ &\leq (1 + |\theta_1| + |\theta_2|)^2 |\theta_1 - \theta_2| \end{aligned}$$

and

$$|h(\theta)| \leq |\theta| + |\theta|^3 \leq 2(1 + |\theta|^3).$$

Hence, Assumption 2 holds with $L = 1, K = r = 2$. On the other hand, one writes

$$\langle \theta_1 - \theta_2, \theta_1|\theta_1|^2 - \theta_2|\theta_2|^2 \rangle = |\theta_1|^2 \cdot |\theta_1 - \theta_2|^2 + (|\theta_1|^2 - |\theta_2|^2) (\langle \theta_1, \theta_2 \rangle - |\theta_2|^2).$$

Since the equivalence of the positions of θ_1 and θ_2 , one also obtains

$$\langle \theta_1 - \theta_2, \theta_1|\theta_1|^2 - \theta_2|\theta_2|^2 \rangle = |\theta_2|^2 \cdot |\theta_1 - \theta_2|^2 - (|\theta_1|^2 - |\theta_2|^2) (\langle \theta_1, \theta_2 \rangle - |\theta_1|^2).$$

Taking the average of the above expression, one gets

$$\begin{aligned} \langle \theta_1 - \theta_2, \theta_1|\theta_1|^2 - \theta_2|\theta_2|^2 \rangle &= \frac{1}{2} (|\theta_1|^2 - |\theta_2|^2)^2 + \frac{1}{2} (|\theta_1|^2 + |\theta_2|^2) |\theta_1 - \theta_2|^2 \\ &\geq \frac{1}{2} (|\theta_1|^2 + |\theta_2|^2) |\theta_1 - \theta_2|^2. \end{aligned}$$

Consequently, Assumption 3 holds with $a = 1/2, b = 0$ and $\bar{r} = 0$. This completes the proof. $\square$

Proof of Proposition 4.3. By the definition of potential U , one easily obtains

$$\begin{aligned} h(x) &= \{6\alpha\tau x_{i,j,k} - \alpha\tau (x_{i+1,j,k} + x_{i,j+1,k} + x_{i,j,k+1} + x_{i-1,j,k} + x_{i,j-1,k} + x_{i,j,k-1}) \\ &\quad + (1 - \tau)x_{i,j,k} + \zeta\tau x_{i,j,k}^3\}_{i,j,k \in \{1,2,\dots,d\}} \end{aligned}$$

for all $x \in \mathbb{R}^{d^3}$. Moreover, for any $x, y \in \mathbb{R}^{d^3}$, by applying $|a^3 - b^3| \leq 2(a^2 + b^2)|a - b|$ for all $a, b \in \mathbb{R}$ and let $c_1, c_2, c_3, c_1^*, c_2^*, c_3^* \in \{1, \dots, d\}$ be such that $|x_{c_1, c_2, c_3}| = \max_{i,j,k} |x_{i,j,k}|$ and $y_{c_1^*, c_2^*, c_3^*} = \max_{i,j,k} |y_{i,j,k}|$, then it yields that

$$\begin{aligned} |h(x) - h(y)| &\leq 12\tau\alpha|x - y| + |1 - \tau| \cdot |x - y| + \tau\zeta \left| \sum_{i,j,k=1}^d (x_{i,j,k}^3 - y_{i,j,k}^3) \right| \\ &\leq (12\tau\alpha + |1 - \tau|) |x - y| + 2\tau\zeta (|x_{c_1, c_2, c_3}|^2 + |y_{c_1^*, c_2^*, c_3^*}|^2) |x - y| \\ &\leq \max\{|1 - \tau| + 12\tau\alpha, 2\tau\zeta\} (1 + |x| + |y|)^2 |x - y|. \end{aligned}$$

At the same time, by the definition of (22), one also obtains for all $x \in \mathbb{R}^{10^3}$

$$|h(x)| \leq (|1 - \tau| + 12\tau\alpha) |x| + \tau\zeta \sum_{i,j,k=1}^d |x_{i,j,k}^3| \leq \max\{|1 - \tau| + 12\tau\alpha, \tau\zeta\} (1 + |x|^3). \quad (79)$$

Thus, the above calculation implies that Assumption 2 holds with

$$L := \max\{|1 - \tau| + 12\tau\alpha, 2\tau\zeta\}, \quad K := \max\{|1 - \tau| + 12\tau\alpha, \tau\zeta\}.$$

Using that $x \mapsto g(x) := \sum_{i,j,k=1}^d |\tilde{\nabla} x_{i,j,k}|^2$ is a convex by composition of convex functions and its gradient $\nabla g(0) = 0$, then one obtains for all $x \in \mathbb{R}^{d^3}$,

$$\langle x, h(x) \rangle = \langle x, h(x) - h(0) \rangle \geq \sum_{i,j,k=1}^d ((1 - \tau)x_{i,j,k}^2 + (\zeta\tau)x_{i,j,k}^4).$$

On the other hand, by the Cauchy-Schwarz inequality, one gets

$$\left(\sum_{i,j,k=1}^d x_{i,j,k}^2 \right)^2 \leq d^3 \sum_{i,j,k=1}^d x_{i,j,k}^4, \quad \langle x, h(x) \rangle \geq \frac{\tau\zeta}{2} \sum_{i,j,k=1}^d x_{i,j,k}^4 \quad (80)$$

for all $x \in \mathbb{R}^{d^3}$ and $|x|^2 \geq 2d^3|1-\tau|/(\tau\zeta) > 0$. Therefore, for all $|x|^2 \geq \max \left\{ 1, \frac{2d^3|1-\tau|}{\tau\zeta} \right\}$, combining (79) and (80), one also gets

$$|x| \cdot |h(x)| \leq (|1-\tau| + 12\tau\alpha) |x|^2 + \tau\zeta |x| \cdot \left| \sum_{i,j,k=1}^d x_{i,j,k}^3 \right| \leq \left\{ \frac{2d^3|1-\tau|}{\tau\zeta} + \frac{24d^3\alpha}{\zeta} + 2d^3 \right\} \langle x, h(x) \rangle,$$

which implies that Assumption 1 holds. Finally, for any $x, y \in \mathbb{R}^{d^3}$, it yields that

$$\begin{aligned} & \langle x - y, h(x) - h(y) \rangle \\ & \geq (1-\tau) \langle x - y, x - y \rangle + (\zeta\tau) \sum_{i,j,k=1}^d (x_{i,j,k} - y_{i,j,k})(x_{i,j,k}^3 - y_{i,j,k}^3) \\ & \geq -|1-\tau| \cdot |x - y|^2 + (\zeta\tau) \sum_{i,j,k=1}^d (x_{i,j,k} - y_{i,j,k}) \{ x_{i,j,k}^2 (x_{i,j,k} - y_{i,j,k}) + (x_{i,j,k}^2 - y_{i,j,k}^2) y_{i,j,k} \} \\ & \geq \frac{\zeta\tau}{2} (|x|^2 + |y|^2) |x - y|^2 - |1-\tau| \cdot |x - y|^2, \end{aligned}$$

where the first inequality holds due to the fact $g(x) = \sum_{i,j,k=1}^d |\tilde{\nabla} x_{i,j,k}^2|^2$ is a convex function and $\nabla g(0) = 0$, and the last inequality holds due to the proof method of Assumption 3 in Proposition 4.2. Therefore, the above inequality leads that Assumption 3 holds with

$$a := \frac{\zeta\tau}{2}, \quad b := |1-\tau|, \quad \bar{r} := 0.$$

Therefore, this completes the proof. $\square$

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(Y. Dai) SCHOOL OF MATHEMATICS AND STATISTICS, HEFEI UNIVERSITY, HEFEI 230601, P.R. CHINA
Email address: daiy@hfu.edu.cn

(J. Huang) SCHOOL OF MATHEMATICS AND STATISTICS, WUHAN UNIVERSITY, WUHAN 430072, P.R. CHINA
Email address: hj_math@whu.edu.cn

(Y.L. Jiao) SCHOOL OF ARTIFICIAL INTELLIGENCE, AND HUBEI KEY LABORATORY OF COMPUTATIONAL SCIENCE, WUHAN UNIVERSITY, WUHAN 430072, P.R. CHINA
Email address: yulingjiaomath@whu.edu.cn

(L.C. Kang) SCHOOL OF ARTIFICIAL INTELLIGENCE, WUHAN UNIVERSITY, WUHAN 430072, P.R. CHINA
Email address: kanglican@whu.edu.cn

(X.L. Lu) SCHOOL OF MATHEMATICS AND STATISTICS, WUHAN UNIVERSITY, WUHAN 430072, P.R. CHINA
Email address: xllv.math@whu.edu.cn